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TOPICAL REVIEW

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Topical Review

Digital image correlation for surface deformation measurement: historical developments, recent advances and future goals

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Abstract

This article is a personal review of the historical developments of digital image correlation (DIC) techniques, together with recent important advances and future goals. The historical developments of DIC techniques over the past 35 years are divided into a foundation-laying phase (1982–1999) and a boom phase (2000 to the present), and are traced by describing some of the milestones that have enabled new and/or better DIC measurements to be made. Important advances made to DIC since 2010 are reviewed, with an emphasis on new insights into the 2D-DIC system, new improvements to the correlation algorithm, and new developments in stereo-DIC systems. A summary of the current state-of-the-art DIC techniques is provided. Some further improvements that are needed and the future goals in the field are also envisioned.

Keywords: 2D-digital image correlation, stereo-digital image correlation, displacement/strain measurement

(Some figures may appear in colour only in the online journal)

1. Introduction

1.1. Fundamentals of DIC

1.1.1. What is DIC and what can DIC do? Digital image correlation (DIC) is a non-contacting image-based optical method for full-field shape, displacement and deformation measurements. DIC techniques first acquire digital images of an object at different loadings (times or states) using digital imaging devices (including but not limited to optical imaging, electronic imaging, and scanning probe imaging facilities), and then perform image analysis with correlation-based (intensity-based or area-based) matching (tracking or registration) algorithms and numerical differentiation approaches to quantitatively extract full-field displacement and strain responses of materials, components, structures or biological tissues [1–10]. Being image-processing techniques in essence, aside from their non-contact, full-field measurement capabilities, DIC techniques offer several distinctive features, such as a simple and inexpensive experimental setup, easy implementation, wide applicability with adjustable spatial and temporal resolutions, and robustness against ambient vibrations as well as ambient light variations. These outstanding advantages, together with the rapid spread of turnkey commercial DIC systems [11–16] in the last two decades, have not only led to the dominance of DIC techniques in the experimental mechanics community over competing interferometric optical techniques, e.g. electronic speckle pattern interferometry (ESPI) and Moiré interferometry (MI), but also engender their prevalence in new areas of application, such as material science, biomechanics, civil engineering, geotechnical engineering, and aerospace engineering.

Theoretically speaking, the image-based DIC techniques can be extended to any test samples, any circumstances and any imaging technologies, provided that the recorded digital images have sufficient and stable intensity variations, and unique correspondences to the physical points on the test sample surface. Indeed, application of this revolutionary optical technique has experienced explosive expansion in the last three decades, ranging from regular metal or polymeric materials to composite materials and biological tissues [17–20], from microscopic or even nanoscale [21–23] to macroscopic scale, with feature size ranging from several micrometers (e.g. fibers) to hundreds of meters (e.g. bridges) or even tens of kilometers (e.g. ground deformation) [24–27], from local surface to full-surface panoramic or interior full-field 3D measurements [28–30], from static or quasi-static loading to high-speed dynamic loading [31–34], and from common laboratory conditions to outdoor conditions or extreme environments (e.g. high-temperature, high-pressure, underwater or radiation) [35–44].

Basically, DIC-measured displacement and strain fields can serve the following four purposes. (1) *Characterization*. The measured kinematics quantities (e.g. displacement, strain, acceleration, strain rate) in response to external loadings directly reveal the mechanical behavior or deformation mechanism of the test materials and structures. (2) *Identification*. Based on the known exerted mechanical or thermal loadings and the measured kinematic fields, various static, dynamic constitutive or structural parameters of the test materials, components and structures (e.g. Young's modulus, Poisson's ratio, coefficient of thermal expansion, stress intensity factor, natural frequency, damping ratio, vibration mode, etc) can be readily determined [45–54], or identified by combining identification methods such as a virtual fields method or FEM updating [55, 56]. (3) *Cross-validation*. These obtained kinematics fields can be used to verify the correctness of theoretical predictions and FEM analyses [57–59]. (4) *Control MTS testing*. By using DIC-measured full-field kinematics, accurate control of various mechanical parameters (including displacement, strain and stress intensity factor history, etc) in a mechanical test performed on a testing machine can be realized, to connect a specific mechanical state with loading configuration or specimen geometry [60–63].

1.1.2. Classification of DIC techniques. Broadly, DIC techniques can be divided into three main categories, as shown in figure 1. A two-dimensional (2D) DIC method using a single fixed camera is limited to in-plane deformation measurement of nominal planar objects. To acquire accurate measurements, some strict requirements on the specimen deformation, loading device and measuring system must be met [7]. If the test object surface is non-planar, or if 3D deformation occurs after loading, the 2D-DIC method is no longer applicable. To overcome the drawbacks of 2D-DIC, a 3D-DIC (also known as stereo-DIC) technique, using two synchronized cameras or a single camera assisted with specially designed light-splitting devices [64], has been developed. Based on the principle of binocular stereovision, stereo-DIC can accurately measure full-field 3D shape and deformation of both planar and curved surfaces, and is thus more useful and practical in real applications. Digital volume correlation (DVC), a direct 3D extension of 2D-DIC, has also been proposed [30, 65]. By comparing the volumetric images acquired by a 3D or volumetric imaging device, DVC emerges as a powerful tool for quantifying the internal deformation of opaque solid objects (e.g. trabecular bone [66], wood [67], rock [68], and cellular foam [69]) and certain semi-transparent biological tissues (e.g. cell [70] and lamina cribrosa [71]).

In addition to the above three well-known and widely adopted DIC techniques, there also exist other forms of DIC techniques with special applicability, as indicated in figure 2. For instance, an off-axis 2D-DIC using a single camera, whose optical axis is oblique to the test sample surface, can measure the deflection (vertical displacement) of an object surface (figure 2(a)) [72]. This technique demonstrates great potential in measuring the deflections of larger engineering structures (e.g. bridges). Besides that, combined with the projection of computer-generated speckle patterns with a digital projector and 2D-DIC (figure 2(b)), the speckle projection method (also known as the shadow speckle method) [73, 74] has also been presented for out-of-plane displacement and shape measurement. Moreover, a digital gradient sensing (DGS) method was proposed [75] for measuring full-field stress gradients in transparent planar solids by combining an elasto-optic effect and 2D-DIC. It uses a camera to observe the speckle target through the specimen (figure 2(c)), and estimates the stress gradient from the DIC-measured displacement fields of the fixed speckle patterns. Subsequently, a reflective-mode DGS method was developed for studying orthogonal slopes and curvatures of optically reflective objects (figure 2(c)) [76]. However, the latter three forms are not as general as the other correlation-based techniques, because they can be used only for special purposes. In order not to make this review too lengthy, here we consider only on 2D-DIC and stereo-DIC techniques that are commonly used for full-field surface deformation measurements. DVC, off-axis DIC, digital speckle projection and DGS techniques are not discussed hereafter.

1.1.3. Implementation procedures of DIC techniques. As illustrated in figure 3, the determination of surface deformation using DIC techniques generally consists of the following three consecutive steps.

(i) ***Speckle Pattern Fabrication.*** The test sample surface must have random intensity variations as a carrier of deformation information [77]. If the sample has evident discernible natural texture, this step can be bypassed. Otherwise, artificial speckle patterns must be decorated on the test sample surface.
(ii) ***Digital Image Acquisition.*** The surface images of the test sample at different configurations are recoded by a single camera (2D-DIC) or by two synchronized cameras (stereo-DIC), and stored in a computer for post-processing. Images of calibration targets should also be

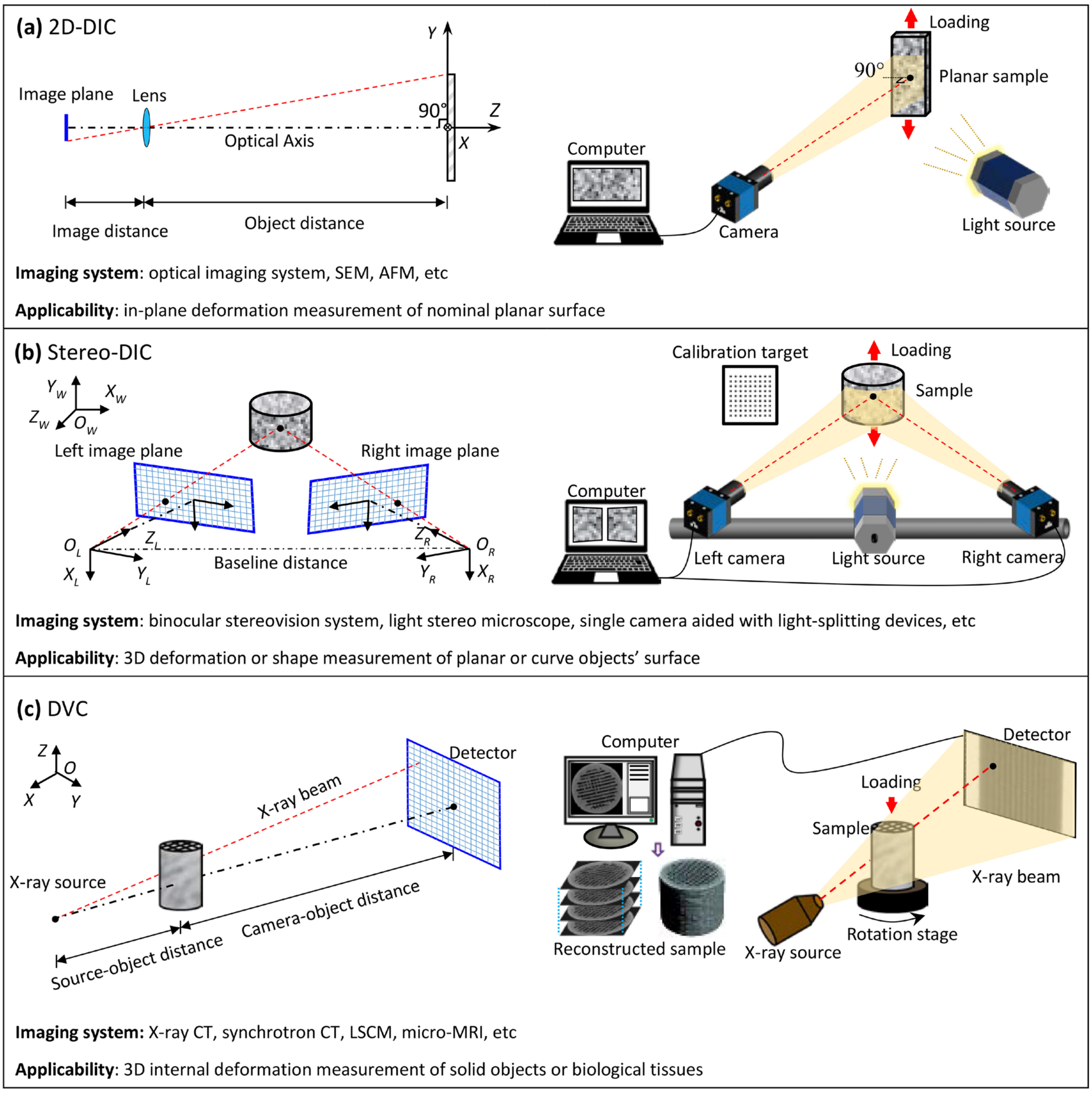

Figure 1. Three typical DIC techniques and their imaging models, imaging systems and applicability.

captured in order to retrieve the magnification factor (2D-DIC) or the intrinsic and extrinsic parameters of the imaging systems (stereo-DIC).

(iii) ***Displacement and Strain Field calculation.*** When using 2D-DIC, full-field displacements (in units of pixels) can be directly retrieved by comparing the digital images of the planar sample recorded at different states, using either subset-based local or FEM-based global correlation-based algorithms [6]. For stereo-DIC, stereovision calibration must be carried out first to determine the intrinsic and extrinsic parameters of the two cameras. Based on the calibrated parameters and disparity maps computed by DIC algorithms, 3D coordinates of all the measurement points can be reconstructed. The subtraction of 3D cloud points before and after deformation yields the surface displacement fields (in units of mm). The obtained displacement fields can be smoothed first and then differentiated or directly differentiated using proper numerical approaches to estimate strain distributions [78–83]. Since the basic concepts, fundamental principles and algorithm details of these DIC techniques have been well documented in various technical papers, review papers and books, they are therefore not repeated here, for brevity.

1.2. Historical developments of DIC

To better indicate the development and popularity of DIC techniques in academic fields, the frequency of publication of three well-known and popular photomechanics methods were retrieved. Figure 4 shows the number of papers retrieved using Web of Science (Science Citation Index Expanded, from 1996 to 2017) and Google Scholar (from 1982 to 2017) by inputting <'digital image correlation'>, <'moire interferometry'> and <'ESPI or DSPI or digital speckle interferometry or

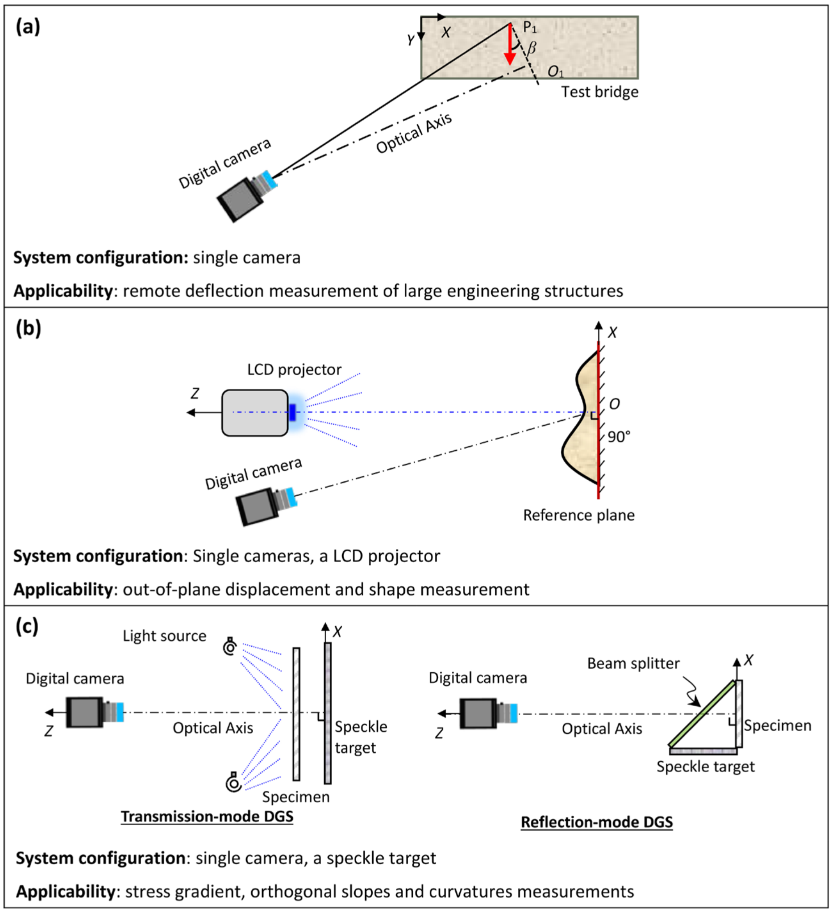

Figure 2. Other forms of DIC-based photomechanics techniques: (a) off-axis DIC, (b) speckle projection method, (c) DGS.

shearography' > in 'topics' respectively. Note that as Google Scholar includes more papers published in many regional journals in other languages, the number of publications is much higher. Nevertheless, these two data sources show the same trends of development of DIC techniques. That is, ESPI and MI seem to be slightly more popular than DIC techniques before 2005. However, DIC techniques have undergone a burst of methodology research and applications since 2005, which are embodied in the exponentially increased number of publications. Presently, it is fair to say that the DIC techniques have been and will continue to be the most popular and most important metrology tool in the experimental mechanics community. Although 2005 appears to have been a watershed year in terms of publication number, here we would like to divide the historical development of DIC into two phases, with the year 2000 as the dividing line: namely, a foundation-laying phase (1982–1999) and a boom phase (2000 to the present). The reason for this division of DIC history is because the theoretical foundations and basic frameworks of DIC techniques were virtually established before 2000.

1.2.1. Foundation-laying phase: 1982–1999. Peters and Ranson, at the University of South Carolina, first proposed the idea of 2D-DIC to the field of experimental mechanics. In a seminal paper published in Optical Engineering in 1982 [1], they tracked the rigid-body motions of the laser speckle patterns scattered from a small aluminum specimen, using a very simple correlation algorithm that combines the simplest cross-correlation (CC) function with the zero-order shape function. Afterwards, in a series of papers [2–5], Sutton and his colleagues continued to improve the original 2D-DIC technique. Specifically, in the two papers published one year later, in 1983 [2, 3], they used random intensities on the sample surface with white light illumination instead of laser speckle patterns to track motion and deformation, adopted a simple least-squares correlation (i.e. sum-of-squared difference, SSD) that considered the homogenous linear deformation (i.e. first-order shape functions) of deformed subsets, and employed an iterative coarse-fine algorithm and bilinear interpolation to minimize the correlation function to determine six deformation terms. In another paper, published in 1985 [4], they further optimized the DIC algorithm by using the more robust normalized cross-correlation (NCC) criterion, polynomial interpolation and a 'two-parameter' iterative algorithm. In 1986 [5], they developed an optimized DIC method which employs the Newton–Raphson (NR) method with differential corrections to minimize NCC. The NR algorithm was demonstrated to be 20 times faster than previous coarse-fine search routines with the same measurement accuracy. Particularly, in 1989 they gave, in great detail, the principles and algorithm implementation of the classic NR algorithm with bicubic spline interpolation for subpixel registration [84]. However, the determination of the Hessian matrix involved in the NR algorithm make its algorithm implementation both complicated

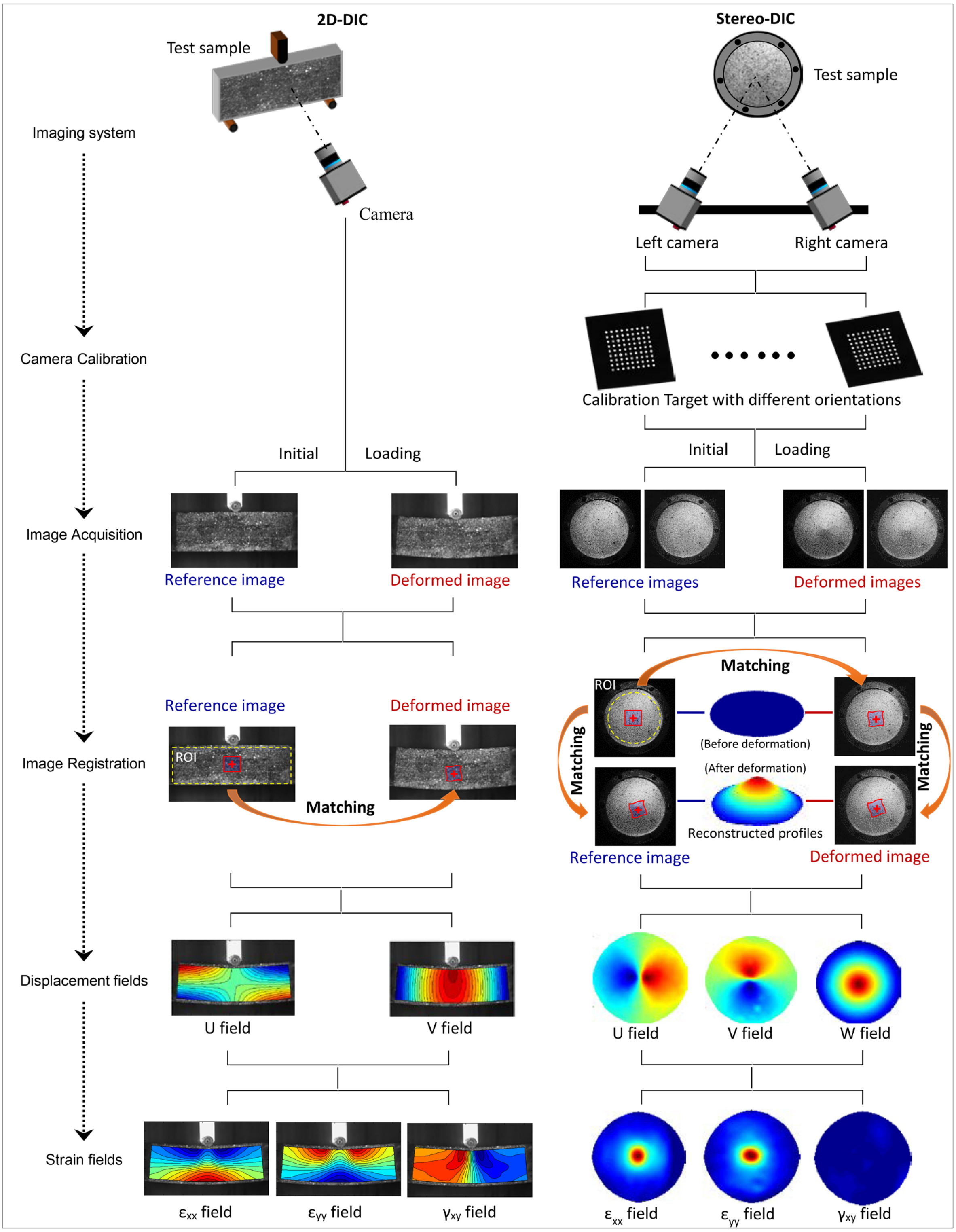

Figure 3. Implementation procedures of 2D-DIC and stereo-DIC techniques.

and computation-intensive. In 1998, Vendroux and Knauss simplified the NR algorithm by using an approximate Hessian matrix, which offered increased computation speed and convergence robustness without loss of registration accuracy [85]. Also, it should be noted that an alternative algorithm for speckle-displacement measurement was proposed by Chen *et al* in 1993 [86]. They used a fast-Fourier transform (FFT) for integer displacement estimation, followed by a peak-finding algorithm based on biparabolic fitting of the correlation map for subpixel registration. Despite being computationally efficient, the peak-finding algorithm was later found to have two evident shortcomings. It cannot deal well with cases of larger rotation and/or deformation, and its subpixel registration accuracy is lower than that of the NR algorithm.

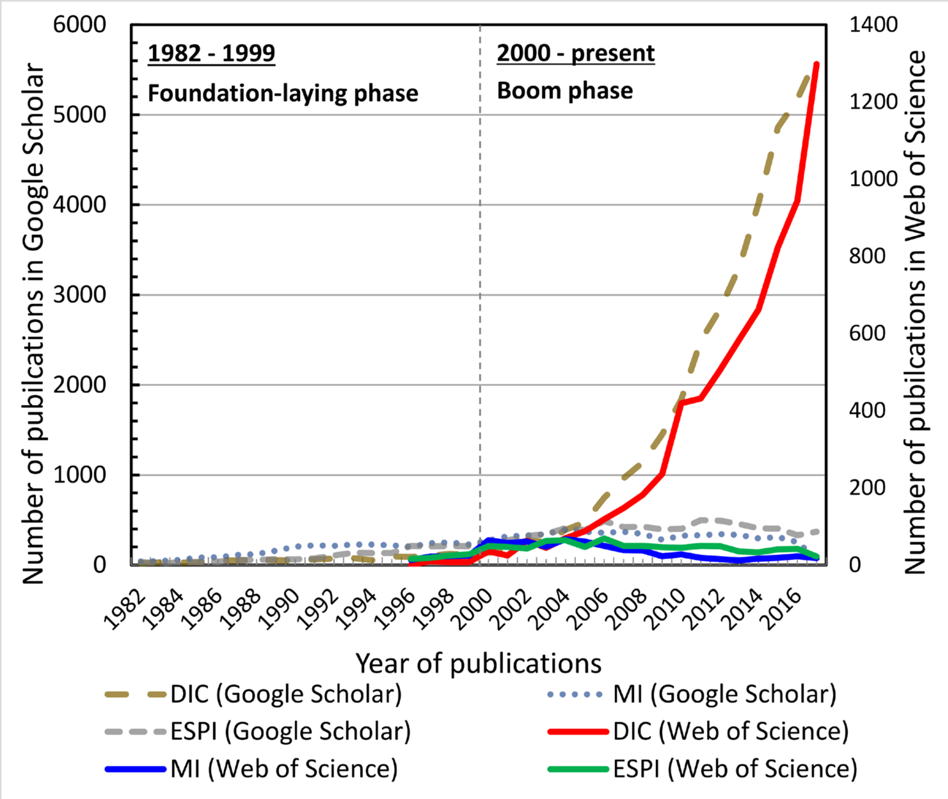

Figure 4. Number of articles involving primary photomechanics techniques since 1982.

During the foundation-laying phase of the DIC technique, other advanced correlation-based experimental techniques were also proposed. For example, to overcome the application limitations of 2D-DIC, and driven by the practical requirements of deformation measurement of non-planar surfaces, stereo-DIC for surface 3D shapes and deformation measurement and DVC for truly full-field internal 3D deformation were proposed. In the early 1990s, by combining the principle of stereovision and 2D-DIC, Luo *et al* [28] developed a two-camera stereovision system for surface deformation measurements. In the late 1990s, to retrieve the 3D displacement and strain maps from 3D volumetric images of a bone specimen generated by high-resolution x-ray computed tomography (x-ray CT), Bay and his colleagues successfully developed DVC for full-field internal deformation measurement of objects with identifiable microstructures, which can be considered as a direct 3D extension of the subset-based 2D-DIC method [30].

In short, during the early foundation-laying phase, pioneers in this field laid the foundation and built the main framework of DIC techniques. The ideas of subset-based correlation methods, including 2D-DIC [1], stereo-DIC [28] and DVC [30], were established in this phase, and have not changed since then. However, DIC techniques in this phase were not as popular as their current prevalence, mainly because their measurement accuracy and precision, computational efficiency, robustness and repeatability were unsatisfactory, for three reasons: (1) the hardware for image capture and image processing were rather expensive, but provided relatively inferior performance (e.g. the image quality and resolution of digital cameras, and the computational capacity of the computers); (2) many details involved in correlation algorithms (speckle pattern quality, subset size, shape function, interpolation methods, calculation path, etc) and their influences on subpixel registration were not fully understood; and (3) the complex error chain of DIC measurements considering the miscellaneous error sources in speckle pattern, imaging process and correlation algorithm and their propagation, were not fully explored.

1.2.2. Boom phase I: 2000–2009. Since 2000, DIC techniques have attracted and mobilized numerous researchers around the world, possibly due to the ever-increasing computational capability of modern computers with falling costs, more easily available imaging devices with better performance, and to their gradually accepted prominent advantages over interferometric optical techniques, together with the widespread acceptance of turnkey commercial DIC systems. Indeed, DIC techniques have undergone explosive growth in publication frequency during the boom phase, with many insightful and highly-cited papers published by researchers from around the world. Following the path paved by pioneers, research efforts in this phase have been dedicated to more accurate, efficient, robust implementations of DIC techniques, with an aspiration to deeper understanding, and better and broader applications of these techniques. In this phase, DIC techniques have developed tremendously due to the contribution of a large number of researchers worldwide, and the developments can be broadly categorized as follows.

1.2.2.1. Introduction of new imaging systems, characterization and correction of errors caused by the equipment used. Aside from regular optical imaging systems, various advanced imaging systems, such as light stereo microscopes [21], fluorescent microscopes [88], scanning probe microscopes (e.g. atomic force microscope (AFM) and scanning tunnel microscope (STM)) [89, 90], electron microscopes (e.g. scanning electron microscope (SEM), transmission electron microscope (TEM) and scanning transmission electron microscope (STEM)) [91–93], high-speed cameras [31], infrared cameras [94] and x-ray cameras [95] were all introduced for deformation measurement at various spatial and temporal scales. The imaging distortions of various imaging systems and their influences on DIC measurements have also received great attention. A radial distortion model and polynomial model were used to approximate the lens distortion of optical imaging systems [32, 96–98]. Drift and spatial distortion were considered in interpreting the imaging errors of scanning probe microscopes and electron microscopes [89–93, 99]. Corresponding approaches based on pre-calibration using in-plane translated image pairs [21], a precisely fabricated grid [98], and rescan tests [99] were proposed to correct the errors due to imaging.

1.2.2.2. Improvement of correlation algorithm. The iterative spatial domain CC algorithms (e.g. the NR algorithm) were continuously refined by considering various algorithm details, such as correlation criterion, gray-level interpolation method, shape function, calculation path and initial guess. Tong [100] evaluated the robustness, reliability and computational cost of four different kinds of correlation criteria using three sets of images, which were of variable brightness, contrast, uneven local lighting and blurring. He recommended the use of the most robust zero-mean normalized sum of squared difference (ZNSSD) criterion. Schreier *et al* [101] examined the position-dependent systematic error (i.e. bias error) of the iterative spatial domain CC caused by gray-level interpolation methods. They revealed that cubic and quintic spline interpolation are preferable to simple polynomial interpolation because of the better convergence and reduced bias errors. Lu *et al* [102] introduced a second-order approximation of the displacement gradients (i.e. second-order shape functions) to approximate local nonlinear deformation occurring to deformed subsets, and claimed that compared with first-order shape function this refinement provides the same accuracy in a small deformation case and better accuracy when a large displacement gradient appears. Soon afterwards, Schreier and Sutton [103] analyzed the systematic errors that arise from the use of an under-matched shape function, i.e. shape functions of a lower order than the actual displacement field. A reliability-guided displacement tracking (RGDT) strategy with automated initial guess predicted from computed neighboring calculated points was proposed by Pan [104], which greatly improved the robustness and efficiency of correlation analysis.

Apart from the iterative spatial domain CC algorithm, various other subpixel registration algorithms, including the iterative, spatial gradient-based subpixel registration algorithm [84], the gradient-based subpixel registration algorithm [105, 106], and the genetic algorithm [107–109], have also been advocated by various researchers in various literatures. In 2006, Pan *et al* compared the performance of three most commonly used subpixel registration algorithms, i.e. the iterative spatial domain CC algorithm, the peak-finding algorithm, and the gradient-based algorithm [110]. The results revealed that the iterative NR algorithm offers higher accuracy, better stability and broader applicability, and is therefore highly recommended for practical use. This research clarified the confusion in selecting the optimal subpixel registration algorithm. Later, a least-squares framework, which uses an iterative least squares (ILS) algorithm for subpixel registration, and a pointwise least squares algorithm for strain estimation, was established by Pan *et al* [82]. It was also proved mathematically that the ILS algorithm is equivalent to the classic NR algorithm, but its fundamental principle and algorithm implementation is much easier and straightforward.

Although the mainstream approach for displacement measurement is the subset-based DIC, which deals with square subsets centered at discrete measurement points individually, alternative approaches for subpixel registration also emerged in the first decade of the 21st century. In particular, by introducing continuous representation of displacement fields using a B-spline function [111] or a finite element formulation [112, 113], a global correlation-based DIC technique can determine the displacements at all calculation points at a time by explicitly ensuring the global displacement continuity and displacement gradients continuity among calculation points. Due to this feature, a global DIC technique can offer seamless link with numerical simulations, and helps to bridge the gap between experiments, modeling and simulation. Furthermore, by incorporating the enrichment strategy of an extended FEM, a global DIC technique can offer accurate measurement of discontinuous displacements encountered in specimens containing cracks [114, 115].

1.2.2.3. Assessment of DIC measurement errors. Real DIC measurement involves a complex and long error chain, which should consider various aspects including out-of-plane motion, speckle pattern, image noise, and correlation algorithm details. Some key factors were singled out and examined separately. For example, Sutton *et al* investigated the out-of-motion effects of a test sample on 2D-DIC and stereo-DIC measurements [116], and convincingly demonstrated the advantages of the latter. Several researchers realized that speckle pattern combined with subset size has an influence on the accuracy and precision of the measured displacements. Possibly based on intuition, several empirical parameters (e.g. mean speckle size, subset entropy) were proposed to assess speckle pattern quality [117, 118]. Theoretical error analyses have also been performed to quantify the influences of image intensity gradients and image noise on the displacements detected by DIC, and revealed that the standard deviation error in measured displacement is proportional to the magnitude of the white noise, but inversely proportional to a novel parameter called the sum of square of subset intensity gradient (SSSIG). The SSSIG was then advocated by Pan *et al* to evaluate the quality of the speckle pattern within individual subsets [119]. Also, by adjusting the SSSIG, a criterion was developed by which to choose a proper subset size for DIC analysis in order to achieve the desired measurement precision.

Real-world and numerical experiments were also performed to assess the performance of DIC techniques by considering more error sources. Haddidi and Belhabeb used in-plane and out-of-plane rigid-body translations to assess the measurement errors of DIC related to lighting, optical lens distortion, the CCD sensor, the out-of-plane displacement, the speckle pattern, step size, subset size, and the correlation algorithm [120]. Wang *et al* theoretically deduced the governing formula of measurement bias and random errors as a function of various factors, including interpolation method, subpixel motion, image noise, image contrast, level of uniaxial normal strain, and subset size [121]. In addition, a collaborative work was carried out by a group of French researchers to assess the metrological performances of various DIC algorithms. By using three synthetic speckle images undergoing sinusoidal displacements with various amplitudes and spatial frequencies, the influences of different calculation parameters (subset size, step size) and various algorithm details (subset shape function, gray-level interpolation, optimization algorithms) on DIC measurement errors were characterized [122].

1.2.2.4. Applications to various fields. In the first decades of the 21st century, the applications of DIC techniques gradually diffused to other fields. Aside from their better application to various branches of solid mechanics (e.g. experimental mechanics, fracture mechanics, fatigue, mechanical testing), DIC has found applications in various natural and artificial materials (including metals [79], foam [17], polymers [48], biological tissue [19], sand [123], wood [124, 125], concrete [126], composite material [18, 127], etc), components (including electronic packages [128], assemblies [129], satellite antennas [130], etc) and structures (including MEMS [131], bridges [25], masonry walls [132], geological structures [133], etc) subjected to various external loadings and experimental environments. Several special applications, including high-speed or dynamic measurement, high-temperature measurement and microscale measurement also began to be present in this period.

It is worth noting that the reader can achieve a much more complete understanding of the full scope of these developments by referring to the excellent book on DIC by Sutton *et al* [6], the very popular review paper on 2D-DIC by Pan *et al* [7], and the technical paper on 3D-DIC with a quite inclusive list of references by Orteu [8], which are almost comprehensive and exhaustive for 2009.

1.3. Aim of this work

Although DIC techniques before 2009 have been well summarized in an excellent book and several highly-cited review papers written by the leading researchers in the fields, the techniques have undergone significant improvements in the last nine years due to the contributions of a large number of researchers around the world. These advances have been making DIC measurements more controllable, robust, versatile, cost-effective, efficient and accurate. The purpose of this paper is to follow the two reviews on 2D-DIC and stereo-DIC that were published in 2009 and to give a personal view of the important developments in 2D-DIC and 3D-DIC since 2010. Given the huge number of publications contributed by researchers worldwide, it is not realistic to cover all the aspects of correlation-based techniques and their numerous applications. Instead, the presentation reflects the author's experiences of and views on certain developments that seemed particularly important, useful and interesting. Lastly, remaining problems and some improvements that are needed are summarized, and the future possibilities in the field are envisioned.

2. Advances in 2D-DIC since 2010

Since 2010, many important advances have been made to various aspects of 2D-DIC, varying from speckle pattern fabrication and quality assessment, new insights in 2D-DIC systems, and new improvements in correlation algorithms. In this section, important advances in these aspects are reviewed, with particular emphases placed on the new insights of 2D-DIC imaging systems, a novel generalized reference sample compensation (RSC) method that makes 2D-DIC measurement more robust and accurate, and the state-of-the-art IC-GN algorithm that has been considered as the de facto standard algorithm in DIC.

2.1. Speckle pattern fabrication and quality assessment

DIC measurements cannot be performed for the specimens without speckle patterns, which act as a carrier of deformation information of the test specimen surface. In addition to their indispensability in DIC measurements, speckle patterns also have important influences on the accuracy and precision of the displacements detected by DIC. Naturally, two questions arise to DIC practitioners. The first question is 'What is a good speckle pattern?', and the second is 'How can we make a good speckle pattern?'

In general, a good speckle pattern should fulfill the following empirical qualitative requirements, as pointed out by various researchers [117, 134–138]: (1) ***High contrast***: with varying grayscale intensities and relatively large intensity gradients; (2) ***Randomness***: a non-periodic and non-repetitive pattern, to facilitate full-field displacement mapping; (3) ***Isotropy***: no obvious directionality in the pattern; (4) ***Stability***: tightly adhere to the sample surface and deform together with the sample surface without evident changes in geometric and grayscale characteristics. To quantitatively evaluate the quality of speckle patterns, a global parameter called the mean intensity gradient (MIG) was proposed by Pan *et al* in 2000 [139], which was extended from the local parameter SSSIG previously defined for quality assessment of a local speckle pattern within individual subsets. Other similar metrics [118, 140–142], including subset entropy [118] and mean subset fluctuation [140], have also been introduced. However, neither of these local and global metrics can be used to assess the randomness in a speckle pattern. Recently, with the purpose of creating optimal DIC patterns, a combined pattern quality metric, which combines SSSIG with a term called the secondary auto-correlation peak height, was proposed by Bomarito *et al* [143]. In this combined pattern quality metric, SSSIG is used as the primary component, since SSSIG was found to be the metric most closely correlated with DIC accuracy, and the secondary auto-correlation peak height is used as a constraint upon SSSIG to ensure that a regular (e.g. checkerboard-type) pattern is not achieved. Further, by introducing a constraint on a local average grayscale to the combined pattern quality metric, Bomarito *et al* [144] developed optimal multiscale patterns that can be used for DIC measurements at multiple magnifications.

To answer the second question above, different speckle pattern types were developed, including natural and artificial speckle patterns. Of course, natural texture features in some materials or structures (soil, wood, rocks, etc) can offer sufficient intensity variations for pattern matching. The inclusions, grain boundaries, additives and the second phases have been considered as natural patterns for micro-scale measurement at higher magnifications of optical microscopy (OM) and SEM. However, in most cases, the speckle pattern should be decorated artificially using either additive or subtractive approaches, as summarized in the review paper by Dong and Pan [77]. Additive approaches for macro samples can be implemented with an airbrush or marker pen if the surface texture of the sample cannot provide sufficient intensity variations. Also, to fabricate speckle patterns at reduced lengths, the focused ion beam (FIB) technique, spin-coating, compressed air technique, e-beam lithography, and vapor-assisted remodeling were proposed, which can make micro- and nano-speckles suitable for high resolution DIC measurement under SEM.

Note that the quality of the speckle patterns generated by most existing techniques is heavily dependent on the operators' experience. To overcome this shortcoming, several techniques, including water transfer printing (WTP) [145], an atomization system [146], and a pre-fabricated speckle pattern stencil [40] have all been proposed, which can produce repeatable DIC patterns with controlled quality. Finally, it is worth noting that, to quantify the accuracy and precision of different DIC codes and provide analysis guidelines to DIC practitioners, synthetic speckle images with precisely applied deformation and well-vetted sample images have been generated and collected to isolate the error sources arising from any given experimental setup and imaging system. These DIC challenge image sets can be downloaded from the website: https://sem.org/dic-challenge/.

2.2. New insights in 2D-DIC imaging system

Although 2D-DIC is simple to implement, its limitation is obvious since it can be used only to determine in-plane deformation of nominal planar surfaces. In most publications regarding 2D-DIC, the ideal pinhole imaging model was implicitly assumed. Recently, it was explicitly pointed out by Pan *et al* [147, 148] that the ideal pinhole imaging model is *neither 'perfect' nor 'stable'* because of the practically existing out-of-plane motion in test planar samples caused by loading and sample deformation, out-of-plane motion in a sensor plane caused by temperature variation of the camera, the unavoidable lens distortion, as well as ambient light variations and ambient vibrations (see figure 5). In fact, the effects of these deleterious issues (out-of-plane motion [116], lens distortion [149], non-perpendicular camera alignment [150], self-heating effect of digital cameras [151], and ambient light variations [152]) on 2D-DIC measurements have been separately investigated by different researchers. However, comprehensively considering all these negative factors in 2D-DIC measurements is essential in order to realize quality and repeatable measurements.

Based on a thorough understanding of the 2D-DIC imaging system, two measures that can be taken to achieve high-accuracy in-plane deformation measurements using 2D-DIC were suggested by Pan *et al* [147]. The first measure is to seek a 'near perfect' and 'ultra-stable' optical imaging system, and the other is to compensate for the displacement errors associated with imperfect and unstable imaging (see figure 5). In 2013, by comparing the performances of three imaging lenses, i.e. a regular imaging lens, an object-side telecentric lens and a high-quality bilateral telecentric lens, Pan *et al* [147] demonstrated that a high-quality bilateral telecentric lens is not only invariant to small changes in both object distance and imaging distance, but also has negligible lens distortion. Thus a 2D-DIC imaging system using a high-quality bilateral telecentric lens can be taken as a *'near-perfect and ultra-stable'* imaging system (figure 6(a)). The system was further improved by introducing the idea of active imaging [153], which was based on the combined use of an actively illuminated monochromatic blue light source and band-pass filters mounted before the imaging lens. Since ambient light variations can be effectively suppressed, active imaging DIC systems using blue illumination demonstrate great robustness against ambient light variations, and thus extend DIC measurements to non-laboratory and extreme high-temperature environments. Using this specially designed 2D-DIC system, high-accuracy stain measurement for re-position tests and accurate determination of viscoelastic Poisson's ratio of solid propellants can be achieved [153, 154]. However, bilateral telecentric lenses are not generally applicable and have several limitations, such as their high price, limited field of view and fixed working distance. It is interesting to note that, with the purpose of better suppressing the thermal radiation emitted from hot objects at a higher range of temperatures, ultraviolet-DIC (UV-DIC) systems [155, 156], which were established based on the combined use of ultraviolet (UV) light, a UV light transmission filter and a UV camera, have also been described. In addition, cross-polarization using orthogonal linear polarizers on light source and camera has also been introduced to DIC systems [157] for higher robustness against direct specular reflections on a sample surface.

To compensate for the errors associated with imperfect and unstable imaging and to realize high-accuracy deformation measurement using a common lens, Pan *et al* proposed a novel RSC method [148]. The RSC method employs a thin, hollow rectangular compensation specimen, which is attached to the testing surface and moves rigidly with the test specimen

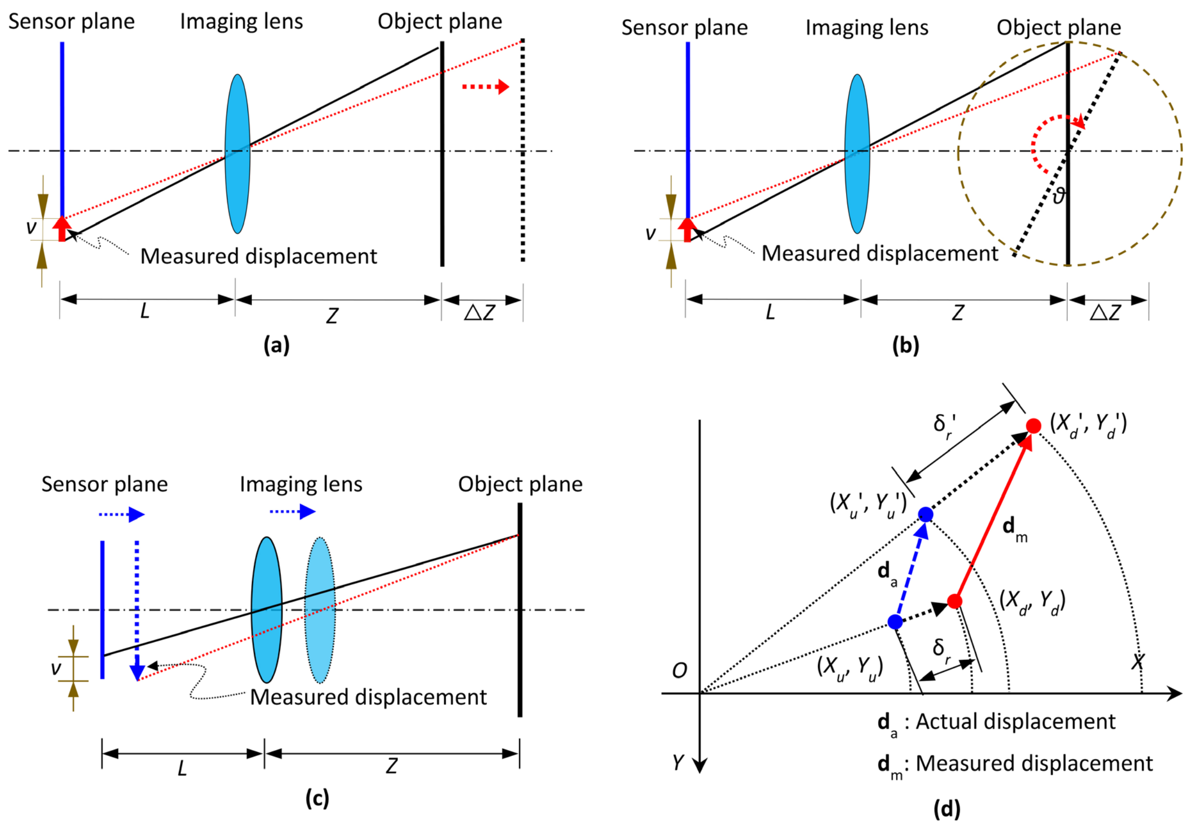

Figure 5. Influence of (a) out-of-plane translation of test object, (b) out-of-plane rotation of test object, (c) self-heating of a camera, and (d) radial lens distortion on the 2D-DIC measurements.

during loading, as shown in figure 6(b). During loading, the region of interest (ROI) and region of compensation (ROC) should be located in the same plane and should always be within the field of view of the camera. In this manner, the displacement and strain errors caused by these detrimental factors (lens distortion, out-of-plane motions in object surface and sensor target) in the ROI and ROC can be represented using the same parametric model. All these error sources are considered and can be effectively eliminated in the proposed RSC method. As such, high-accuracy 2D-DIC measurements can be realized using low-cost common imaging lenses, or even a camera phone [158].

Benefiting from these recent advances, various high-accuracy video extensometers that can deliver accurate, real-time strain measurements have been proposed, as shown in figure 7. Specifically, by combining active imaging and a bilateral telecentric lens with the state-of-the-art DIC algorithm, an advanced video extensometer was proposed by Pan *et al* [159] for real-time, high-accuracy strain measurement in mechanical testing. High-accuracy video extensometers based on the use of a simplified RSC method were also developed [160]. With the aid of reference samples attached to the test sample, virtual strains caused by out-of-plane motions and lens distortion can be effectively compensated in a real-time manner. In addition, several other techniques were also introduced into 2D-DIC-based video extensometers to minimize or compensate the virtual strains due to out-of-plane motions [161–163]. For instance, Hoult *et al* [161] presented a correction scheme by using two cameras to view both sides of the loaded specimen to improve the strain measurement accuracy of the 2D-DIC technique. Inspired by Hoult's idea, Zhu *et al* [162] proposed an easy-to-implement high-accuracy optical extensometer by using a single camera and a dual-reflector imaging technique. This technique can be considered as using two virtual cameras to record the front and rear surfaces of a specimen. By averaging the two strain results of the front and rear optical extensometers, the virtual strains due to out-of-plane motion, non-perpendicularity between the camera optical axis and specimen surface, as well as the initial bending of the specimen, can all be automatically eliminated.

2.3. Subset-based correlation algorithm

After recording the digital images of a test object before and after deformation, the deformed images can be compared against the reference image to extract full-field displacements and strains. Generally, the ROI, subset size, and grid step should be first specified in the reference image to determine the exact positions of calculation points. Then the subset-based DIC aims to track the motion of each point of interest individually in the deformed images. To achieve efficient, accurate and robust subset matching, an objective function combining a similarity measure (i.e. correlation criterion) and a proper shape function must be defined first. The established objective function is then optimized using a subpixel registration algorithm and certain intensity interpolation method to determine the desired deformation vector.

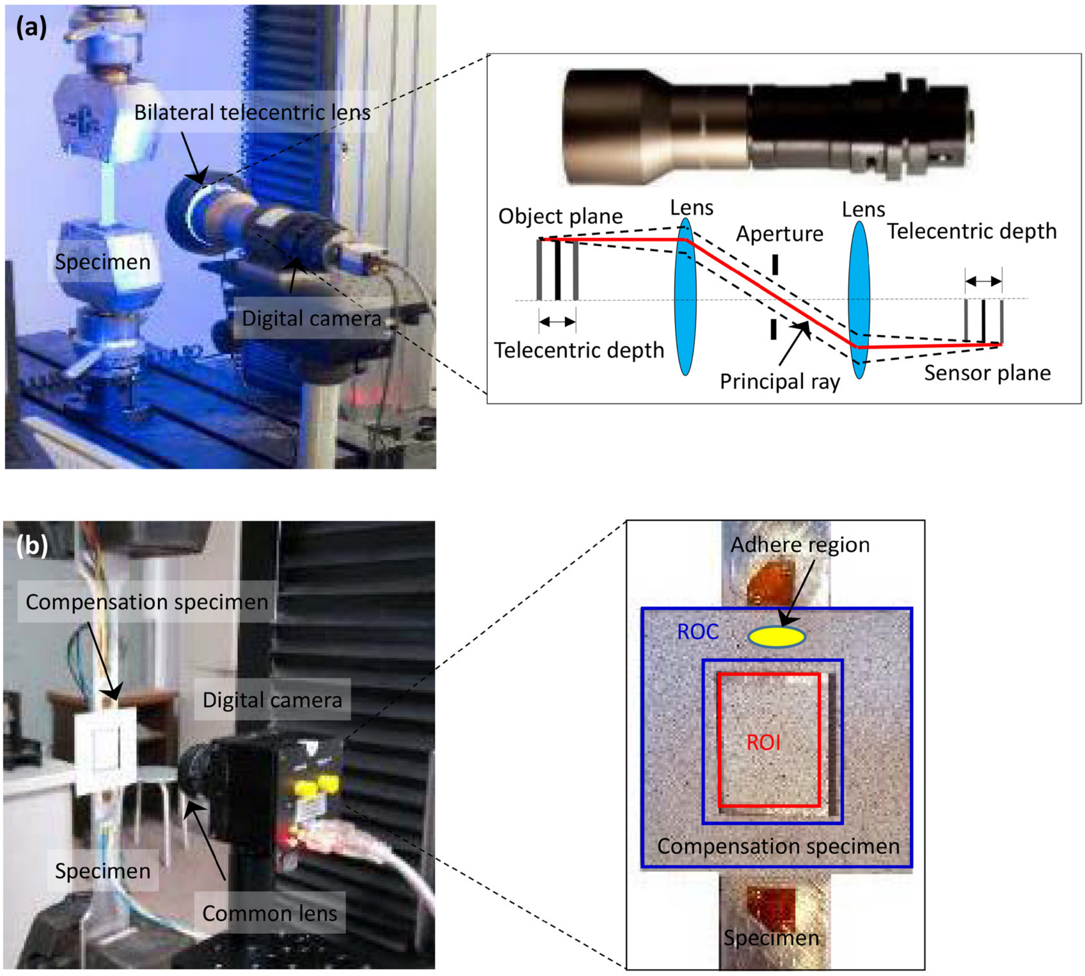

Figure 6. (a) 'Near perfect and ultra-stable' 2D-DIC system using a high-quality bilateral telecentric lens with active imaging; (b) high-accuracy 2D-DIC measurement based on the RSC method.

2.3.1. Correlation criterion. The correlation criterion is defined to quantify the degree of similarity (or difference) between the reference subset and its deformed counterpart, and thus is of fundamental importance in DIC. In 2010, various correlation criteria used in the literature are classified into four categories according to their mathematical definitions: i.e. cross-correlation (CC), sum of absolute difference (SAD), sum of squared difference (SSD), and parametric sum of squared difference (PSSD) [164]. In practice, the intensity changes between images recorded at different configurations may be induced by various reasons. Thus, a robust correlation criterion should be used to accommodate the intensity variations in the deformed images. Otherwise, significant displacement measurement errors may occur due to the mismatch of intensity change model. Based on this consideration, a zero-mean normalized cross-correlation (ZNCC) criterion, a ZNSSD criterion, and a $PSSD_{ab}$ criterion with two additional unknown parameters have been highly recommended for practical use, as they are insensitive to the potential scale and offset changes of the target subset intensity. Also, theoretical analyses performed by Pan *et al* [164] indicate that these three correlation criteria are actually equivalent, which elegantly unifies these correlation criteria for pattern matching. For brevity, mathematical definitions of three types of most commonly used correlation criteria and their performances are given in table 1. The transversal relationship among the three highly recommended correlation criteria is also provided.

In addition to the correlation criterion summarized in [9, 164], several modified correlation functions have also been developed in recent years. To deal with possible nonlinear intensity changes in target subsets, Liu *et al* [165] presented a $PSSD_{abc}$ criterion, in which the nonlinear intensity change model incorporates three unknown parameters with a, b, c denoting the offset, scale, and second-order intensity changes, respectively. Also, considering the fact that intensity gradients of an image are less sensitive to illumination variations, Xu *et al* [166] introduced an intensity gradient-based SSD criterion instead of the intensity-based ZNSSD criterion, to deal with nonlinear illumination variations. Both these two correlation criteria can be used in certain special cases where obvious nonlinear intensity changes occurred between the reference and target images, due to non-uniform illumination or other reasons. In addition, a weighted ZNSSD (WZNSSD) criterion was proposed by Huang *et al* [167], by introducing a weighted Gaussian window to the classic ZNSSD criterion to self-adaptively adjust subset size during correlation analysis.

2.3.2. Under-matched or overmatched shape functions. In using a subset-based DIC, a proper shape function must be

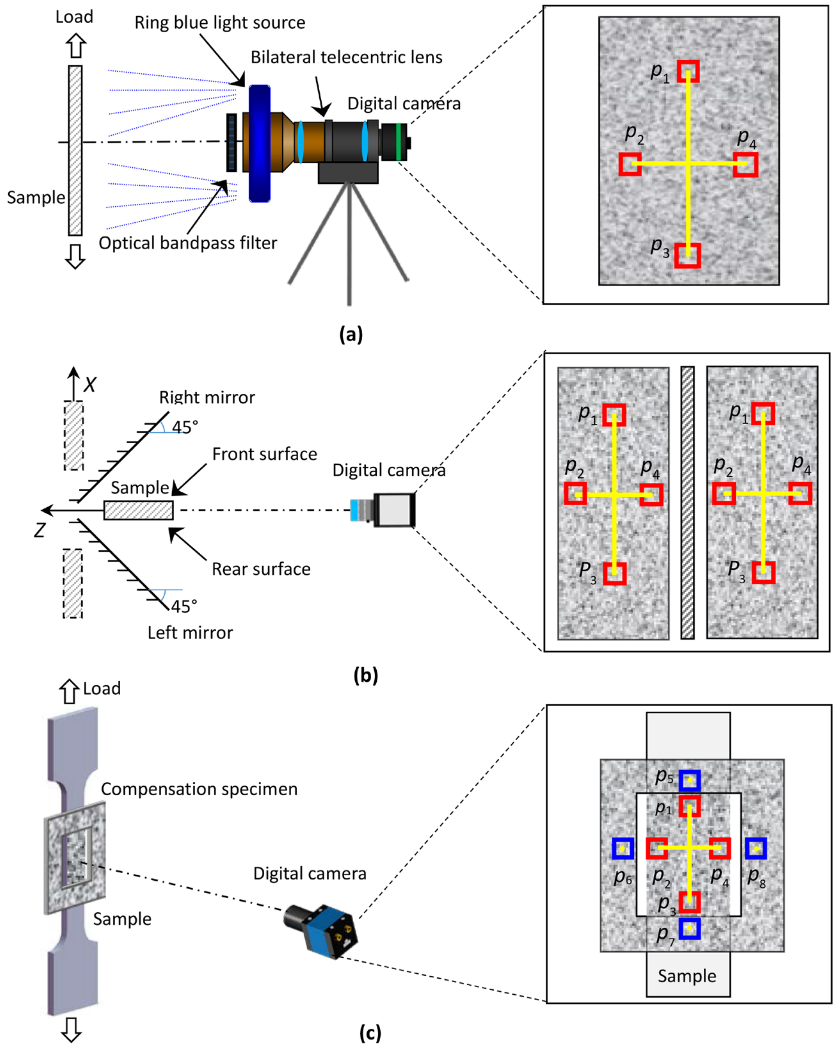

Figure 7. Three kinds of high-accuracy video extensometer based on 2D-DIC: (a) the video extensometer using a bilateral telecentric lens; (b) the video extensometer using dual-reflector imaging; and (c) the video extensometer using a simplified RSC method.

Table 1. Three types of correlation criteria used in DIC for evaluating the similarity degree between the reference subset and the deformed subset.

Performance	CC criteria	SSD criteria	PSSD criteria
Sensitive to all changes of the deformed subset intensity	$C_{\mathrm{CC}} = \sum f_i g_i$	$C_{\mathrm{SSD}} = \sum (f_i - g_i)^2$	
Insensitive to offset changes of the deformed subset intensity	$C_{\mathrm{ZCC}} = \sum (\bar{f}_i \bar{g}_i)$	$C_{\mathrm{ZSSD}} = \sum [\bar{f}_i - \bar{g}_i]^2$	$C_{\mathrm{PSSD_b}} = \sum (f_i + b - g_i)^2$
Insensitive to scale changes of the deformed subset intensity	$C_{\mathrm{NCC}} = \frac{\sum f_i g_i}{\sqrt{\sum f_i^2 \sum g_i^2}}$	$C_{\mathrm{NSSD}} = \sum \left(\frac{f_i}{\sqrt{\sum f_i^2}} - \frac{g_i}{\sqrt{\sum g_i^2}} \right)^2$	$C_{\mathrm{PSSD_a}} = \sum (af_i - g_i)^2$
Insensitive to the scale and offset changes of the deformed subset intensity	$C_{\mathrm{ZNCC}} = \frac{\sum \bar{f}_i \bar{g}_i}{\sqrt{\sum \bar{f}_i^2 \sum \bar{g}_i^2}}$	$C_{\mathrm{ZNSSD}} = \sum \left(\frac{\bar{f}_i}{\sqrt{\sum \bar{f}_i^2}} - \frac{\bar{g}_i}{\sqrt{\sum \bar{g}_i^2}} \right)^2 = 2\left(1 - C_{\mathrm{ZNCC}}\right)$	$C_{\mathrm{PSSD_{ab}}} = \sum (af_i + b - g_i)^2 = \sum \bar{g}_i^2 \left(1 - C_{\mathrm{ZNCC}}^2\right)$

Here, $\bar{f} = \frac{1}{n}\sum_{i=1}^{n} f_i$, $\bar{g} = \frac{1}{n}\sum_{i=1}^{n} g_i$, $\bar{f}_i = f_i - \bar{f}$, $\bar{g}_i = g_i - \bar{g}$ with f_i, g_i denoting the intensity value of the ith pixel point within the reference and target subsets.

chosen to approximate the underlying displacement field of the target subsets, for accurate subset matching. Shape functions with varying orders of Taylor's expansions (e.g. zero-, first- and second-order) have been proposed. However, since the actual deformation occurring in the deformed subsets cannot be known *a priori* in most practical measurements, mismatch problems (under-matched or over-matched) inevitably arise, which lead to additional errors in the measured displacements. For clarity, the theoretically derived systematic errors and random errors due to mismatched shape functions are summarized in figure 8. Specifically, following the pioneering work by Schreier *et al* [103], Xu *et al* [169] theoretically studied the

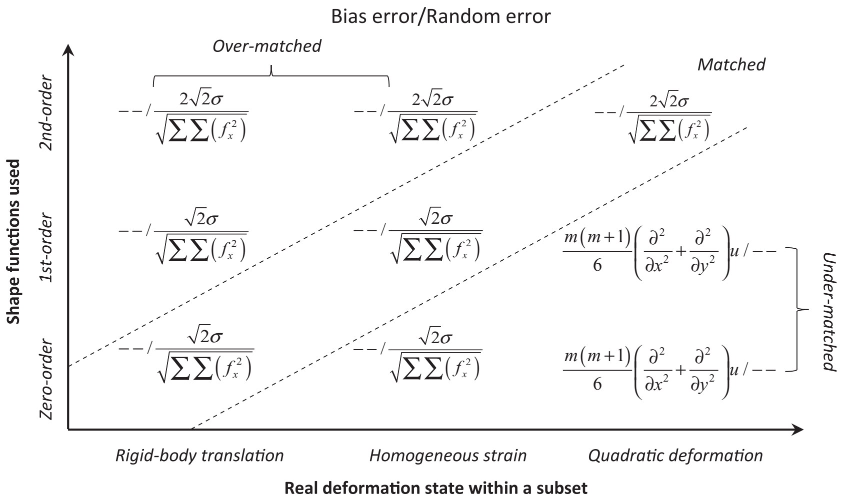

Figure 8. Dominant bias errors and random errors for matched and mismatched shape functions. ('——' indicates that the corresponding bias error or random error can be neglected. For matched and over-matched shape functions, it is assumed that the bias error can be suppressed using high-order intensity interpolation methods or low-pass Gaussian pre-filtering.)

systematic errors in local deformation due to under-matched shape functions, and pointed out that an under-matched systematic error in strain is directly related to subset size, grid step, strain window size, and second-order displacement gradient. In addition, the displacement errors associated with over-matched shape functions were examined experimentally and theoretically by Pan's group [170, 171]. It is shown that over-matched shape functions do not bring additional systematic error but induce increased random error. To be specific, the use of second-order shape functions is able to effectively reduce the possible systematic error in using first-order shape functions [171], while the possibly increased random errors can be suppressed by using a relatively larger subset predicted from the noise level of the speckle image and the SSSIG of the interrogated subsets. Thus, it can be explicitly claimed that second-order shape functions, capable of depicting more complex local deformation, should be used for practical DIC applications as a default.

These regularly used first and second-order shape functions can well represent continuous displacement fields, but cannot characterize displacement discontinuity caused by cracks, crevices, dislocations or shear bands. In these cases, a standard subset-based DIC generally delivers poor results. To overcome this limitation, Poissant and Barthelat proposed a 'subset splitting' procedure into a conventional subset-based DIC on discontinuous fields [172], which allows the splitting of subsets into two sections when a discontinuity is detected. Valle *et al* and Hassan *et al* introduced the Heaviside function used in extended-FEM [173, 174] and Burgers vectors [175] employed for dislocation description, respectively, to model the discontinuities. The correlation criterion using modified subset functions enriched with additional variables can be solved with the classic NR iteration algorithm to detect displacements. Sousa *et al* [176] proposed a pointwise DIC coupling CC with spatio-temporal differential techniques for assessing discontinuous displacement fields. In addition to these approaches, post-processing techniques based on the difference in displacements between the four points [177] or phase congruency [178] have also been proposed to detect the positions of discontinuity and the discontinuous displacement fields.

2.3.3. Subpixel registration algorithms and intensity interpolation methods. As mentioned earlier, Pan *et al* proved that the NR algorithm offers the highest subpixel registration accuracy and best stability [110]. However, the heavy computation complexity of the classic NR was considered to be a big challenge. After carefully examining the implementation of the classic NR algorithm, two redundant calculations (i.e. integer displacement searching and subpixel intensity interpolation) were identified and then eliminated by Pan *et al* in 2010 [179]. Specifically, by introducing a RGDT strategy and an intensity interpolation lookup-table approach, Pan and Li developed a fast DIC using the NR algorithm, which is about 100–200 times faster than the classic version, without any sacrifice in registration accuracy.

To further eliminate the redundant computations involved in the classic NR algorithm without sacrificing its subpixel registration accuracy, Pan *et al* proposed a more efficient inverse-compositional Gauss–Newton (IC-GN) algorithm for fast, robust and accurate full-field displacement measurement [180] (see figure 9). The developed IC-GN algorithm, free of re-evaluating the Hessian matrix and reconstructing subpixel intensity gradients in each iteration, generates similar results to but is about 3–5 times faster than the FA-NR algorithm. After its formal advent in the DIC community, IC-GN has been a research hotspot and continually refined by a variety of researchers [181–183]. For instance, Gao *et al* [181] introduced second-order shape functions to the IC-GN algorithm to improve the accuracy of non-uniform deformation measurement. Although the equivalence of the classic NR algorithm with forward additive mapping strategy and the recently introduced IC-GN algorithm has been proved in existing studies [184, 185], practical implementation of the two subpixel registration algorithms does involve differences, and therefore leads to different performance. Shao *et al* demonstrated

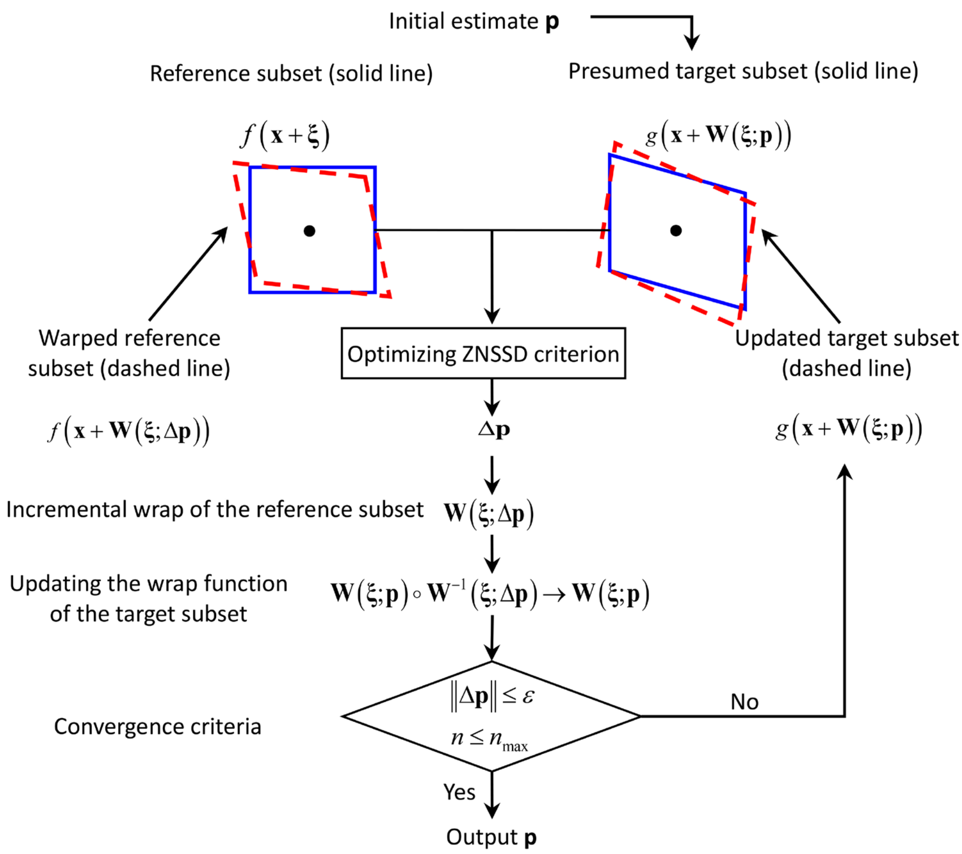

Figure 9. Schematic illustration of the principle of inverse-compositional Gauss–Newton algorithm.

that the IC-GN algorithm has better noise robustness than the FA-NR algorithm [182], because no noise-induced bias is presented in the IC-GN algorithm if the gray gradient operator is chosen properly. Pan and Wang carried out detailed theoretical error analyses of the two algorithms, and derived analytic formulae that can quantify both the bias error (systematic error) and the variability (random error) of the IC-GN and FA-NR algorithms with four different interpolation methods [183]. The results also confirmed that the IC-GN algorithm leads to reduced bias error in displacement estimation by eliminating noise-induced bias error, and gives rise on average to smaller random errors in displacement estimation in cases of high noise levels or when using small subsets. Because of the enhanced subpixel registration accuracy, better noise-proof performance, and higher computational efficiency, the IC-GN algorithm has been strongly recommended as a standard subpixel registration algorithm for practical DIC applications. As a matter of fact, the IC-GN algorithm has been openly used in an open-source DIC package (Ncorr) [186], and adopted as an improved image-based deformation measurement technique for geotechnical applications instead of PIV [187]. Recently, the IC-GN algorithm has also been extended to DVC for accurate and efficient sub-voxel registration of volumetric images [188–190].

During implementation of a subpixel registration algorithm, a certain interpolation scheme should be adopted to reconstruct the intensity at mapped subpixel locations in the deformed images. Since Luu *et al* [191] employed a family of recursive interpolation schemes based on B-spline representation to enhance the accuracy of 2D-DIC analysis, B-spline interpolation kernels have been widely adopted in practical DIC applications. Afterwards, optimized B-spline-based recursive filter [192, 193] as well as 4-, 6- and 8-tap B-spline interpolation methods were successively introduced and became prevalent in most DIC codes for higher subpixel interpolation precision. Although high-accuracy interpolation schemes ensure reduced errors in measured displacement, these errors still prevent DIC from taking a better deformation measurement. Accordingly, following Schreier's classic work in 2000 [101], Su *et al* [194] and Baldi *et al* [195] respectively investigated the errors due to various interpolation schemes from theoretical and experimental perspectives. Furthermore, numerous efforts have been made to reduce and eliminate the interpolation-induced errors in DIC by introducing diverse image pre-filtering techniques [196, 197], random subset offset strategy [198] and a self-correction scheme based on its anti-symmetric feature [199]. Among these solutions, a Gaussian pre-filtering technique prior to correlation analysis presented by Pan in 2013 [196] is both effective and easy to implement, and thus has been widely employed in DIC applications and commercial DIC packages.

2.3.4. Initial guess, convergence conditions and calculation path. Being local iterative optimization algorithms, both the classic NR algorithm and the state-of-the-art IC-GN algorithm require an initial guess of deformation sufficiently close to the true value to initialize the iterative calculation. Indeed, the accuracy of the estimated initial guess directly affects the convergence characteristics of these nonlinear optimization algorithms. Recently, some novel algorithms commonly used in computer vision, such as genetic algorithms [200], scale invariant feature transform (SIFT) [201, 202], and Fourier–Mellin transform (FMT) [203], have been introduced to provide an initial guess to the NR or IC-GN algorithms. These initial guess algorithms, although being computation-intensive, generally demonstrate extraordinary performance, insensitive to large rotations and deformations. Also, to deal with deformed images with serious decorrelation due to large deformation, serious illumination fluctuations, or other reasons, Pan *et al* [204] proposed an incremental DIC method which uses an automatic reference image updating scheme to update the reference images based on the estimated correlation coefficient of seed points. For deformed images with serious decorrelation, incremental displacements extracted by comparing the current deformed image and the updated reference image are cumulated to determine the overall deformation.

Also, NR or IC-GN algorithms require a preset convergence criterion to terminate iterative calculation. Basically, stringent convergence criteria lead to an increased number of iterations and lessen the computational efficiency, whereas convergence conditions that are too loose enhance the computational efficiency but may cause a loss of registration accuracy. Pan investigated the effect of various convergence criteria on the efficiency and accuracy of the IC-GN algorithm in terms of convergence speed and radius of convergence using real experimental images [205]. Recommendations were also given for the selection of proper convergence criteria for more efficient implementation of the IC-GN algorithm.

The calculation path is also a key issue that should be considered in DIC to retrieve full-field displacements. Path-independent analysis, which has been explicitly used from the inception of DIC, deals with each calculation point independently, without an initial guess predicted from neighboring calculated points. It generally consists of two steps, i.e. integer displacement searching using a CC algorithm implemented either in the spatial-domain or frequency domain with FFT. Recently, improvements have been made to accelerate conventional path-independent DIC calculation strategies. Specifically, sum-up tables were introduced by Huang *et al* [206] to avoid redundant calculation involved in spatial-domain integer-pixel displacement searching. Jiang *et al* introduced a fast Fourier transform-based cross-correlation (FFT-CC) algorithm for initial guess estimation with integer-pixel accuracy [207]. Obviously, path-independent DIC analysis can be speeded up by high-performance parallel computing with the aid of GPU [208]. However, it cannot deal with images with inter-frame large rotation and/or deformation. To increase the robustness and efficiency of conventional path-dependent DIC analysis, a robust RGDT strategy presented by Pan in 2009 [104] has been highly recommended. Guided by the ZNCC coefficients of computed points, RGDT can choose an optimal calculation path, while predicting a complete initial guess to the adjacent calculation points. Zhou *et al* [209] refined the initial guess propagation function in using RGDT to provide a more accurate initial estimate and speeded-up convergence of iterative calculation by considering more deformation components (e.g. displacement gradients). However, the path-dependent RGDT strategy can only be implemented point by point, using serial computing. Thus, it not only limits the computational efficiency of DIC analysis, but also wastes the powerful parallel computation capability of modern computers with multicores. To remedy this weakness, Pan *et al* proposed [210] an improved RGDT strategy, which not only maintains the robustness of the existing RGDT strategy, but can be implemented using multi-thread parallel computation to dramatically enhance the computational efficiency of DIC analysis.

2.4. FE-based correlation algorithm

Since 2010, FE-based global DIC has attained increasing attention and has been refined continuously. Advances were mainly made to the following aspects.

(1) ***Accuracy improvement***. Réthoré *et al* introduced a Non-Uniform Unique B-spline (NURBS) shape function in 2010 [211] to reduce the uncertainty of displacement and displacement gradients when using FE-based DIC. Ma *et al* then established an 8-node-quadrangular FE-based DIC (Q8-DIC) in 2012 [212] to ensure higher accuracy for heterogeneous deformation measurement and higher flexibility for specimens with circular edges. For better accuracy in using FE-based DIC, some important issues, including correlation criterion, bias reduction, initial guess and convergence condition, were well addressed by Wang *et al* in 2015 [213]. More importantly, to overcome the compromise between spatial resolution and uncertainty, a series of important research work was carried out in recent years. By introducing multi-grid formulation and the Tychonoff regularization, and Fedele *et al* presented a novel variational formulation for FE-based DIC in 2013 [214] to enhance resolution and accuracy of full-field kinematic measurements. Likewise, in order to ensure the robustness of displacement results in the case of very small element sizes, Yang proposed a global DIC method combined with Tikhonov regularization in 2014 [215] by automatically selecting regularized parameters via an L-curve method. Recently, Passieux *et al* adopted a multiscale approach in FE-based DIC to improve the local measurement resolution and bridge the measurement at near-field and far-field scales [216].

(2) ***Efficiency improvement***. Passieux *et al* developed a global PGD-DIC in 2012 [217] to avoid the main drawback (i.e. the computational cost) of standard FE-based DIC techniques without losing their advantages (e.g. modularity, continuity). Recently, Mortazavi *et al* combined an improved spectral approach (ISA) in FE-based DIC to reconstruct continuous displacement fields based on the Fourier decomposition for an accurate, time- and memory-efficient heterogeneous measurement [218].

(3) ***Adaptability improvement***. Wang and Ma developed an h-adaptive scheme for Q8-DIC [219], in which the optimal element size can be adaptively selected for different deformation regions by refining the mesh, to allow more accurate measurement for complex heterogeneous

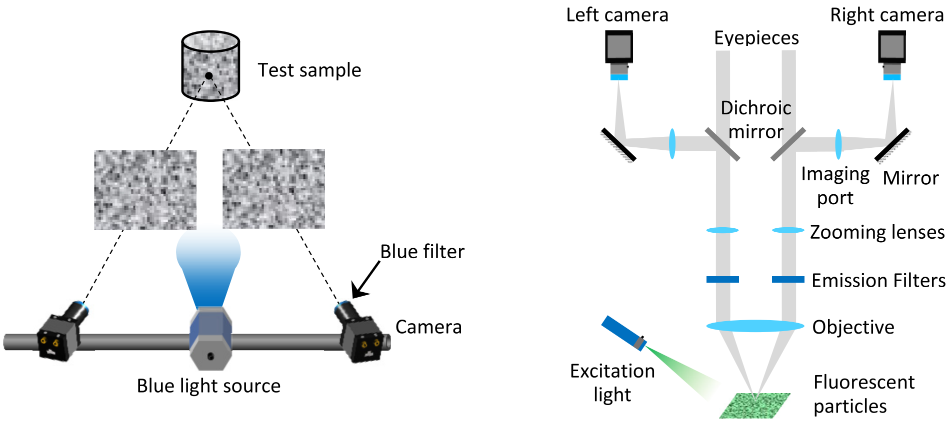

Figure 10. (a) An optimized stereo-DIC system combing monochromatic blue illumination with band-pass filter; (b) microscopic stereo-DIC system based on fluorescent stereo microscope.

deformation. Wittevrongel *et al* proposed a *p*-adaptive global DIC method in 2015 [220] to adaptively add nodes to the pre-defined elements so that the elements have sufficient degrees of freedom to characterize the large non-uniform deformation of the local area (such as plastic deformation, stress concentration, etc).

Subset-based local DIC and FE-based global DIC are the two primary image matching methods that have been extensively investigated and regularly used in the DIC community. Subset-based local DIC has been used in almost all commercial DIC packages because of its straightforward implementation and high efficiency. However, some researchers argued that FE-based global DIC can offer better measurement accuracy due to the enforced continuity between element nodes [221]. A comprehensive comparison between local and global DIC approaches had been confusing and unclear since the appearance of global DIC. Recently, a detailed performance comparison between these different DIC methodologies in terms of measurement accuracy and computational efficiency was reported by Pan's group [222, 223]. Local DIC shows advantages over global DIC in the following several aspects.

(1) For homogeneous deformation, local DIC using a first-order shape function is slightly superior to global DIC in terms of displacement precision, and Q8-DIC with doubled element size is advantageous over Q4-DIC. For heterogeneous deformation, local DIC using a second-order shape function and global Q8-DIC, which can describe more complex deformation, are preferable, but the former performs better in this case.
(2) Local DIC shows much higher computational efficiency than global DIC, since the latter necessitates complex large matrix operations and more iterations under the same convergence condition. Specifically, local DIC performs approximately 5–25 times faster, depending on the specific calculation parameters used.
(3) Local DIC would outperform global DIC in analyzing deformed images with local decorrelation areas that are due to large deformation or damage of the sample and significant local intensity changes [37]. In these cases, by using a robust RGDT strategy, local DIC can identify and isolate these decorrelated points without affecting the displacement tracking of other points. By contrast, the flexibility of global DIC with enforced global continuity is questionable in these cases.

Despite this, global DIC does have several merits in certain specific cases. (1) By using a mesh compatible with regular FEM analysis, global DIC can be directly linked with numerical simulations, thus not only helping to bridge gaps between experiments, modeling and simulations, but also contributing to the minimizing of the discretization error by comparing measured and simulated displacements in a unique FE basis. (2) Benefiting from its global consideration and inter-element continuity constraint, global DIC allows a smaller element size, down to several pixels, in contrast to local DIC, in which the subset size must be large enough to contain sufficient variations in gray levels to warrant accurate matching. Thus, the possibility of using a very small element in global DIC makes it more appropriate to measure greatly heterogeneous deformation.

3. Advances in stereo-DIC since 2010

Since 2010, many research efforts have been dedicated to stereo-DIC, to make this practical deformation measuring technique more robust, versatile and accurate. In this section, a number of significant developments made to stereo-DIC systems are first reviewed, including conventional binocular stereo-DIC, single camera pseudo stereo-DIC, and multi-camera stereo-DIC systems. Advances in stereovision calibration and stereo-correlation algorithm are then discussed. Also, several works dedicated to error analysis and uncertainty quantification of stereo-DIC are outlined.

3.1. New developments in stereo-DIC systems

3.1.1. Conventional binocular stereo-DIC system. To further extend the scope of stereo-DIC to more challenging environments (outdoor or high-temperature environments), several improvements have been devoted to enhancing the robustness of conventional stereo-DIC systems. Pan *et al*

[224] developed an optimized stereo-DIC system, as shown in figure 10(a), which combines monochromatic blue illumination with band-pass filters. Because ambient variations and thermal radiation of a hot object can be effectively suppressed, accurate shape and 3D deformation measurements are available in non-laboratory conditions or extremely high-temperature environments. By using this kind of optimized stereo-DIC system, several groups successfully realized high-temperature deformation measurement of samples subjected to infrared heating [36–39, 225] or laser heating [226]. To suppress specular reflection, polarized light and polarized filters were introduced to a conventional stereo-DIC system to realize precise shape measurement of a coin [227]. It is worth noting that, by slightly changing the optical configuration of existing stereo-DIC, a special stereo-DIC [228] was proposed for accuracy-enhanced constitutive parameter identification using a virtual field method (VFM). In this configuration, the optical axis of the left camera is perpendicular to the test planar sample. Since it can deliver accurate strain measurements without being affected by out-of-plane movements, the special stereo-DIC is promising to be a standard measuring tool for VFM identification.

Instead of using common digital cameras and regular optical lenses, new imaging devices and optical systems have also been introduced to realize 3D surface shape and deformation measurements for various purposes. Nguyen *et al* [229] employed infrared (IR) cameras in two Kinect sensors and one infrared-radiation projector for accurate measurements of 3D shape, motion and vibration. Since projected invisible IR speckle patterns were used for image matching, this technique can be used for textureless objects. However, this technique cannot detect accurate displacement components of an object without decorated speckle patterns. Chen *et al* [230] developed a telecentric stereo micro-vision system along with an accurate calibration technique for shape and deformation measurement of limited fields of view. Additionally, Hu *et al* established a microscopic stereo-DIC system [231–233] using fluorescent stereo microscopy (figure 10(b)), which allows the acquisition of 3D surface profilometry and deformation at microscale and submicron scales. The fluorescent imaging can effectively resolve the issues of specular reflection owing to moisture or smooth shiny surfaces. This significant advantage facilitates the successful measurements of microscopic 3D surface profilometry and deformation of biofilm [231–233], as well as the thickness and profile of transparent material [232]. To realize deformation and profile measurements at the micro-scale, quantitative stereovision in a SEM was realized by Zhu *et al* [234]. This was achieved by capturing duplicate images of the sample surface after tilting the SEM stage, and modifying distortion removal techniques developed for 2D SEM-DIC. A similar approach called multiscale stereo-photogrammetry [235] was also developed to reconstruct, visualize and analyze complete 3D fracture surfaces from SEM images, which offers better insight into fracture surfaces at different levels of detail and deeper comprehension of fracture mechanisms. Also, to meet the increasing demands on full-field high-speed 3D deformation measurements, a high-speed stereo-DIC (HS-stereo-DIC) technique using two synchronized high-speed or ultra-high-speed cameras has been developed and widely explored. In recent years, the applications of HS-stereo-DIC have been reported in a large variety of issues, including measurements of full-field vibration [52, 53], deformation responses to explosive blast loading underwater [41, 236] and under free air [237, 238], and experimental investigations of the shock response of materials [239–241].

3.1.2. Single-camera pseudo stereo-DIC system. In conventional stereo-DIC systems, the requirement for two synchronized cameras usually makes the construction of a regular binocular stereo-DIC system both expensive and complicated, especially in high-speed applications using two high-speed or ultra-high-speed cameras. For this reason, single-camera stereo-DIC techniques have attracted great attention in the last few years and were comprehensively reviewed by Pan *et al* [64]. These techniques can dramatically reduce the investment in hardware and eliminate the need of precise camera synchronization. Generally speaking, to realize single-camera stereo-DIC measurements, the simplest and most commonly used approaches can be implemented by placing an additional optical device (e.g. a diffraction grating [242–244], a bi-prism [245–247] or a four-mirror adapter [248, 249]) in front of the existing imaging lens, as shown in figure 11. With the aid of these light-splitting elements, two or more views of the test object's surface via different optical paths can be simultaneously imaged onto left and right halves of the camera sensor.

However, when two or more views of the object are separately imaged on non-overlapping regions of the sensor, the effective area of the ROI is reduced, resulting in the spatial resolution being halved or even less. To overcome this shortcoming, Yu and Pan proposed a novel single-camera stereo-DIC method using a single 3CCD [250] or CMOS [251] color camera and a skillfully-designed color separation device, as is shown in figure 11(d), which takes full advantage of the spatial resolution of the camera sensor. In this method, the rays reflected by the object surface pass through two optical paths with two different optical band-pass filters installed and then image on the blue and red channels of the sensors. Thus, an overlapped two-channel image of the object surface is recorded. By comparing the blue and red channel sub-images using the regular stereo-DIC algorithm, the full-field shape and 3D deformation information of the object surface can be retrieved. This novel single-color camera stereo-DIC technique inherits all the merits of the standard stereo-DIC, and significantly reduces the hardware investment and technical complexity. Therefore, it is expected to have great potential in high-speed applications, such as impact engineering and vibration tests.

3.1.3. Multi-camera stereo-DIC system. The binocular and single-camera stereo-DIC enable 3D shape, motion and deformation measurements of planar and curved objects to be taken, by using image-pairs of an object's surface rendered from two diverse views. However, these methods may still face difficulties in the following two circumstances. Firstly, for objects of large physical size, the use of stereo-DIC systems may be limited by the camera resolution or the required

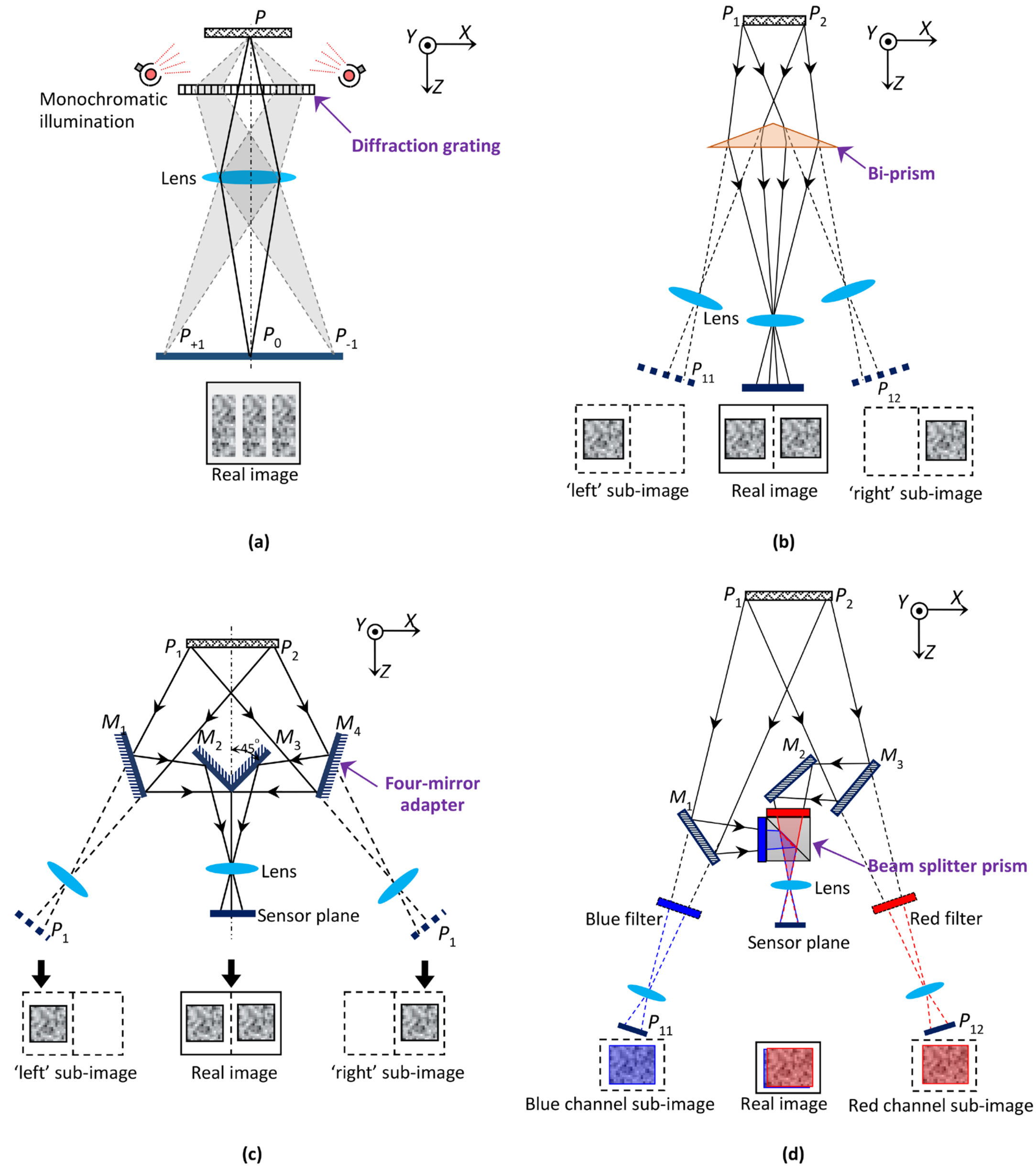

Figure 11. Optical arrangements of four different single-camera stereo-DIC systems: (a) diffraction-based single-camera stereo-DIC system, (b) bi-prism based single-camera stereo-DIC system, (c) single-camera stereo-DIC with a four-mirror adapter, and (d) single camera stereo-DIC system using a 3CCD camera.

working distance in covering the FOV of interest. Secondly, complex curvature of a test object may produce a dead zone that cannot be covered by both views. To solve these challenges, multi-camera stereo-DIC systems, which can record the surface images of the test object from three or more different views with three or more cameras, have been developed in recent years. According to the placement of the cameras, Orteu *et al* [252] divided multi-camera systems into two main configurations: (1) a 'surrounding cameras' configuration for measuring the whole object, and (2) a 'wall-of-cameras' configuration for measuring large quasi-planar objects. By pairing these cameras as several regular stereo-DIC systems, the 3D shape and deformation at different regions of an object's surface are first measured individually by each stereo-DIC system. Then, the results obtained by each individual stereo-DIC system are converted, stitched, or mapped to a universal coordinate system with pre-calibration parameters or with the aid of special makers.

Figures 12(a)–(d) illustrate the specific configuration of typical multi-camera stereo-DIC systems. In 2011, Orteu *et al* [252] developed a four-camera stereo-DIC system for full-field shape and 3D displacement measurements. As illustrated in figure 12(a), one of these cameras is specified as the master camera, and the others are paired with it individually as conventional stereo-DIC systems. The requirement of a common FOV between the master camera and the other cameras can guarantee a more accurate measurement, but in the meantime, the effective measurement region is constrained. Later, Wang *et al* [253] described a multi-camera stereo-DIC system through pairing two adjacent cameras as stereo-DIC systems,

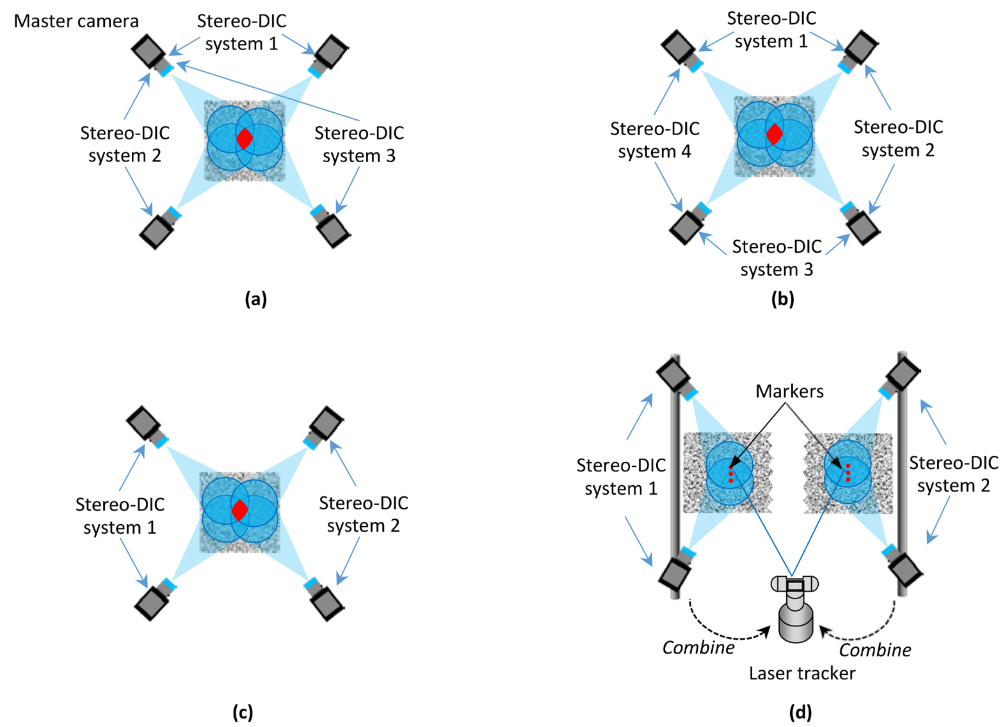

Figure 12. Conventional multi-camera stereo-DIC systems with three or more cameras: (a) a master camera paired with remaining cameras (see Orteu *et al* 2011 [252]); (b) each camera pairs with its two adjacent cameras (see Wang *et al* 2013 [253]); (c) all cameras are divided into several separated pairs (see Chen *et al* 2014 [254] and Chen *et al* 2013 [255]); and (d) several distributed stereo-DIC systems (see Malowany *et al* 2017 [259]).

as shown in figure 12(b). A common FOV of only two adjacent cameras is required, rather than of all cameras, and this approach therefore has a larger measurement region than that of [252]. The geometries and deformation fields from different stereo-DIC systems are then stitched together by coordinate transformations. A multi-camera stereo-DIC system can also be implemented by combining several separated (or pairwise) stereo-DIC systems (figure 12(c)) [254, 255], in which neighboring pairs have a 10% overlapped FOV covered by both cameras. All cameras can be calibrated simultaneously to allow these cameras to be considered as a single system, and therefore the whole surface measurement can be implemented without a stitching operation. Thanks to the whole surface measurement capability, this kind of multi-camera stereo-DIC system has been used to measure whole-field thickness strain and thinning limit [256]. This was realized by simultaneously measuring the front and rear surfaces of a specimen, and then combining the measured results to a universal coordinate system with the aid of a specially designed double-sided calibration plate. In some industrial applications, the regions to be measured are distributed, and there is no overlapping FOV between any two stereo-DIC systems. To stitch together spatial data from different stereo-DIC systems, Malesa *et al* [257, 258] proposed a technique using geodetic surveying, while Malowany [259] utilized a laser tracker to follow makers and to estimate transformations between stereo-DIC systems (see figure 12(d)).

Aside from the above-mentioned conventional configurations, pseudo multi-camera stereo-DIC systems that use only a single binocular stereo-DIC system or a single camera have also been reported for cost reduction and space saving. Leblance *et al* [260] used a portable stereo-DIC system to measure the full-field deformation of a large scale wind turbine blade. As is shown in figure 13(a), the deformation of multiple separated FOVs are obtained using the system at different times and are then stitched together using reference points covered by two neighboring FOVs. Spera *et al* [261] and Genovese *et al* [262] proposed a more compact and inexpensive solution for 360-degree shape and deformation measurements. As is shown in figure 13(b), a single camera is rotated around the test object and a specially-designed cylindrical calibration target, which are presented as the zoomed view at the left of figure 13(b). Or, conversely, the test object and calibration target rotate while the camera is stationary [263]. In the case that the sample has dimensions of around 1 mm, the aforementioned methods are no longer applicable. Genovese *et al* [264] developed a panoramic stereo-DIC, equipped with a single camera and a conical mirror. The camera views the test sample from above, through the reflection of a mirror. By slightly tilting the mirror, they were able to capture a few images from different views, which is equivalent to a pseudo multi-camera stereo-DIC system. These pseudo multi-camera stereo-DIC systems using a single stereo-DIC system or a single camera can greatly reduce the necessary investment. However, it is worth noting that they are practicable only for quasi-static conditions. Very recently, a mirror-assisted panoramic stereo-DIC system was proposed by Chen and Pan [265] that can overcome this limitation and measure full-surface 360-degree deformation even under continuous loading in material testing. As is shown in figure 13(c), this system comprises a single binocular stereo-DIC system

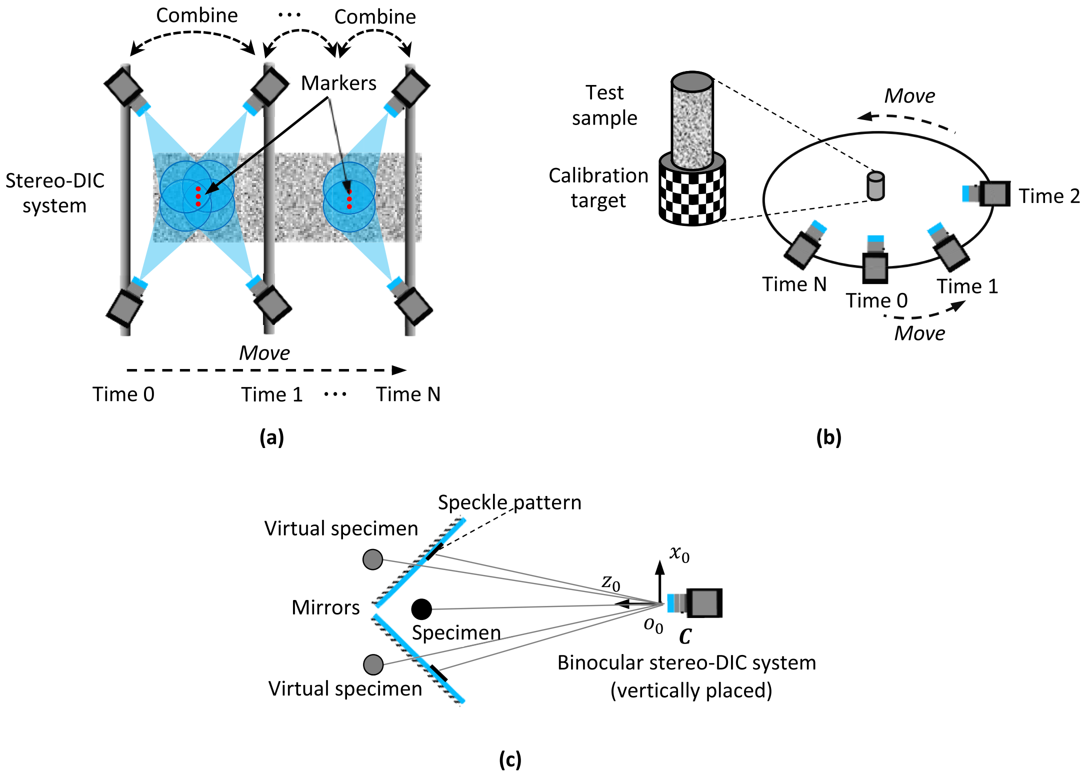

Figure 13. Pseudo multi-camera stereo-DIC systems with just one or two cameras: (a) a stereo-DIC system moves to cover the whole surface (see Leblanc *et al* 2012 [260]); (b) a single camera moves around the test sample (see Genovese *et al* 2016 [262]), and; (c) a mirror-assisted pseudo multi-camera stereo-DIC system (Chen and Pan 2018, to be published [265]).

and two mirrors mounted behind a specimen. With the help of the reflection of the mirrors, different segments of the specimen's full surface are captured and reconstructed by the stereo-DIC system. All the measurements can then be merged into a global system by utilizing the speckle patterns pre-made onto the mirrors.

3.2. Stereovision calibration and stereo-correlation algorithm

3.2.1. Stereovision calibration. To achieve high quality stereo-DIC measurements, accurate calibration of the stereo rig (i.e. the two-camera unit for capturing images simultaneously) is the first and most essential step. Camera calibration provides both the intrinsic (center point, lens distortion coefficients, focal length of each camera) and extrinsic parameters (translation and rotation between the two cameras) necessary for triangulation. In the field of computer vision, the procedures for camera calibration can be roughly classified into two categories: photogrammetric calibration and self-calibration [266]. The former uses certain calibration targets with known precise geometry, while only images of a static scene from different perspectives are required in self-calibration [267–269]. Camera calibration itself has been a very active research topic and witnessed its long developing history. Numerous advances have been made in this field, among which the early works, such as the direct linear transformation method [270] and the two-stage method [271] established the foundation of camera calibration. Due to the high accuracy and easy implementation, the landmark technique [266] using a 2D planar pattern, proposed by Zhang, has become the *de facto* standard method for calibration. Virtually all of the stereo-DIC techniques, from self-developed to commercial stereo-DIC system, are based on this technique.

To achieve higher stereo-DIC measurement accuracy, a slight improvement in calibration accuracy is appreciable. In 2013, Reu [272] carefully examined the influences of various details (e.g. number of calibration images, calibration target size, image contrast and noise) involved in stereo calibration on the calibration uncertainty. From this investigation, it was found that accurate determination of various parameters involved in the calibration is a prerequisite for accurate stereo-DIC measurement. As one of the most fundamental and foremost steps, the location of control points plays a decisive role in camera calibration. By adopting circular control points instead of square patterns in a checkboard calibration target, and using a first order elliptic approximation to take the perspective effect into account, camera calibration results with a reprojection root-mean-squared-error (RMSE) of around 0.05 pixels can be achieved with the bundle adjustment approach. Recently, Vo *et al* [273] proposed an advanced geometric camera calibration technique, which employs a frontal image concept in conjunction with a hyper-precise control point detection scheme based on DIC. This technique can further reduce the reprojection RMSE to below 0.01 pixels.

To realize the deformation measurement of large engineering components or structures with an FOV of interest of several square meters or even larger, regular photogrammetric calibration may be impractical due to the need of an extremely large calibration board. To address this challenge, Shao *et al* [274] adopted a self-calibration method by combining speckle analysis with classical relative orientation algorithm. The intrinsic parameters of both cameras are calibrated in the laboratory in advance. Subsequently, the corresponding points in image pairs matched using DIC are used for rotation and translation estimation between cameras. Although the scale information is undetermined, the technique is sufficient for strain measurement. If incorporated in a single-lens 3D extensometer [275], the speckle-based self-calibration method also allows for real-time strain measurement based on automatic system calibration using a reference speckle image of the specimen.

3.2.2. Stereo-correlation algorithm. In stereo-DIC, the correspondence of the projection of physical points in left and right images must be established using correlation algorithms. Together with the calibrated parameters of the stereo rig, 3D coordinates of measurement points can be recovered with the classic triangulation method. Further, by subtracting the 3D coordinates of the same physical points at the initial state from those of deformed states, three displacement components caused by external loading can therefore be computed. Correlation analysis in stereo-DIC can be separated into two steps, namely stereo matching and temporal matching (or tracking). The goal of the stereo matching is to precisely register the same physical points in the two images captured by the left and right cameras, while temporal matching is to track the same physical points in the consecutive images recorded by the same (left or right) camera at different times or states. Temporal matching can be directly implemented using the well-established subset-based 2D-DIC algorithm. Stereo matching is more challenging due to the large perspective distortion between the images captured by different cameras, and this task is commonly considered to be the most difficult part of stereovision measurement. Particularly, in order to warrant accurate and efficient surface 3D deformation measurements, some issues, such as the matching strategy, correlation algorithm, shape function and initial guess, must be handled with care in stereo matching.

Due to the non-linear nature of the perspective projection, the use of affine shape functions in stereo matching generally leads to substantial errors for large subset sizes and large stereo angles. For this reason, second-order shape functions should always be used in stereo matching. With the purpose of realizing high-accuracy stereo matching with high efficiency, Gao *et al* [181] developed an improved IC-GN algorithm using second-order shape functions (IC-GN^2) in 2014. For temporal matching, the IC-GN algorithm with first-order shape functions is still preferable in most cases, owing to its higher computational efficiency. Based on the IC-GN algorithm and a seed point-based parallel method for parallel computation, stereo-DIC [276] analyses can be implemented in real-time.

To enable fast and accurate stereo matching when images from different cameras have undergone large deformations due to a large stereo angle, feature-based matching using the famous scale-invariant feature transformation (SIFT) algorithm has been introduced to estimate an initial guess for subset-based matching algorithms (e.g. the NR algorithm or IC-GN algorithm). The SIFT algorithm is very robust and can cope well with the scaling, translation, rotation, distortion (affine transformation), occlusion and illumination-variation issues encountered in real image pairs. Also, to deal with serious decorrelation presented in image pairs suffering large deformation, Tang *et al* [277] proposed a large deformation measurement scheme for stereo-DIC by combining an improved coarse search method with a reshaped reference subset and a reference image updating scheme. Recently, Genovese *et al* [278] proposed an image-morphing method and a shape-morphing method. Both methods generate a series of synthetic images to simulate a gradual deformation, which are then used for DIC matching with an incremental approach.

In general, three different matching strategies (figure 14) can be adopted for stereo-DIC. The first strategy, shown in figure 14(a), matches the left reference image to the right reference image (i.e. stereo matching only for the initial state), and all the left and right deformed images are matched to their initial states (temporal matching). For the second strategy, shown in figure 14(b), all the deformed left and right images are correlated to the left reference image. As for the last matching strategy, depicted in figure 14(c), all the left images after deformation are compared to the left reference image, using temporal matching, and then each corresponding right image is compared to the current left image with stereo matching. As discussed above, second-order shape functions should always be adopted in stereo matching, which means stereo matching is much more computationally intensive than temporal matching. Since the first strategy performs stereo matching only once, it is considered to be the best choice in practical measurements. Also, as indicated in figure 14(a), using the first matching strategy, reference subsets should be updated in the right reference image, which generally assumes subpixel locations and should be interpolated. It is worth noting that the computational points in the reference image of the right image series are always at subpixel locations. To eliminate the additional systematic error due to intensity interpolation during reference image updating, Zhou *et al* [198] proposed an adaptive subset offset scheme, in which the reference subsets in the updated reference images are translated to the nearest integer positions, rather than being interpolated at subpixel locations. This simple scheme not only increases computational efficiency by avoiding subpixel interpolation, but also effectively eliminates the periodic systematic errors. Recently, the adaptive subset offset approach [190] has been introduced to DVC to realize interior large deformation measurement with enhanced accuracy and efficiency.

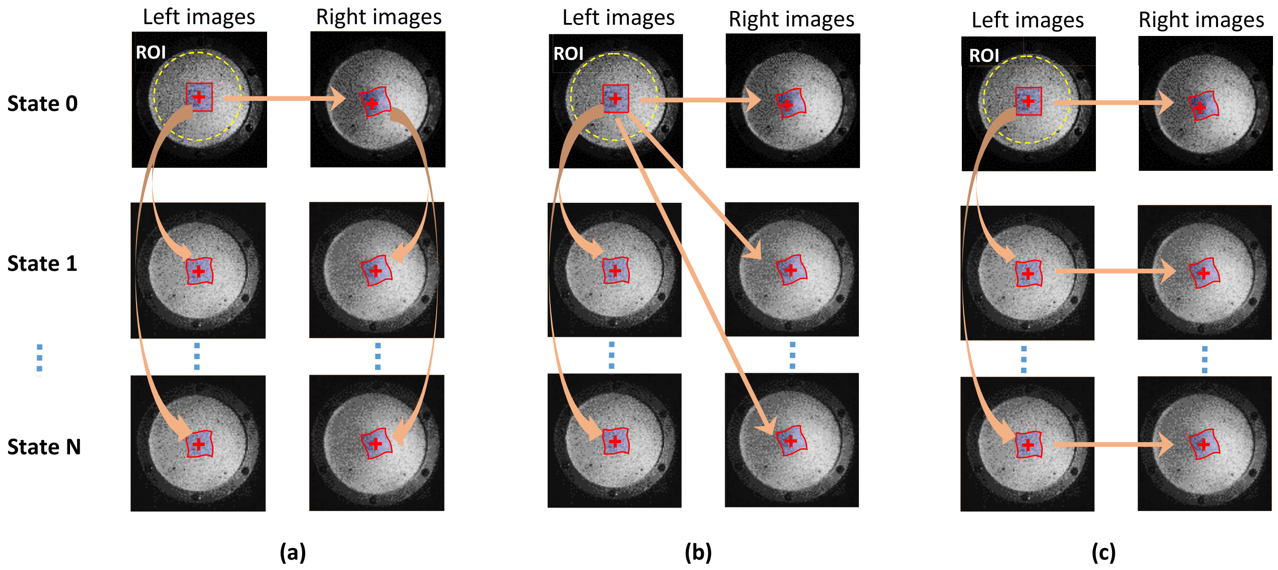

Figure 14. Three major strategies for the stereo-correlations in stereo-DIC: (a) stereo matching for initial left and right images, and temporal matching for subsequent left and right image series; (b) temporal matching for left image series and stereo matching for all the right images using the original left image as the reference image, and; (c) temporal matching for the left images and stereo matching for all the right images, using the current left image as the reference image.

3.3. Error analysis and uncertainty quantification for stereo-DIC

Stereo-DIC is now a standard measurement technique in widespread use; hence, it is important to understand its measurement errors and uncertainties in real experiments. However, compared with 2D-DIC, it is more difficult to evaluate the errors involved in stereo-DIC measurements, because additional complexity and uncertainties involved in stereo rig calibration and correlation can be propagated to final results. But while even theoretical analysis of the error of stereo-DIC is almost impossible, some simplified or statistical methods can be attempted. Wang *et al* [279] established error propagation equations based on the principle of stereo-DIC for quantitative error assessment. Once the camera parameters are determined, one can predict both expectation and variance of 3D positions using the derived error propagation equations. Some simplified simulations and actual experiments [279] were performed, and confirmed the effectiveness of the developed theoretical formulae for variance estimation in both 3D positions and surface strains. Also, based on the error propagation equations, the technique developed by Hu *et al* [280] is available for practical stereo-DIC application rather than simplified configuration. Besides the errors of the conventional two-camera stereo-DIC system, with the flourishing of single-camera stereo-DIC in engineering, the error propagation equations are also used for the uncertainty determination of single-camera stereo-DIC. Yu and Pan [249] made a theoretical study of the single-camera stereo-DIC measurement uncertainty caused by various structural parameters. Based on some reasonable simplification, theoretical uncertainty formulae were established and then validated by a series of real experiments. The results of the experiments demonstrated that the uncertainty will increase non-linearly with the increase of virtual pan angle, and uncertainty in W-displacement is significantly larger than that of the rest. As with the error analyses of many complicated systems, the Monte Carlo method was used by Reu [272] to estimate the stereo-DIC uncertainty. Numerous calibrations were implemented to estimate the uncertainty of calibration, which was further propagated to the uncertainty of 3D positions and motion using Monte Carlo. Using the proposed method, the influence on the 3D position uncertainty of various parameters, including all intrinsic and extrinsic parameters, number of calibration images, calibration target size and calibration target dot spacing, etc, were thoroughly investigated.

The miscellaneous and complicated error sources involved in stereo-DIC measurements make a comprehensive error analysis difficult or even impossible, and thus Balcaen *et al* [281, 282] investigated different error sources, separately, using simulated images. Compared with images from physical experiments, simulated images generated from a finite element displacement field are better for error analysis because different errors can be analyzed separately. They first explored the calibration quality [281], and found that at least 50 image-pairs of the calibration target, with well-known spacing between dots and covering the entire FOV, is highly recommended for stereo-DIC calibration. Various error sources [282], including triangulation errors, numerical errors, interpolation biases, etc, were also activated and compared with a reference state which is free of all errors. A series of simulated experiments demonstrated that a higher stereo angle yields a better precision in depth direction, but at the cost of in-plane measurement accuracy, which conformed with the conclusion demonstrated in [279]. These works also confirmed that good calibration is a prerequisite for accuracy improvement of stereo-DIC measurements.

Apart from the calibration and correlation errors, temperature change will also degrade stereo-DIC measurements but is always neglected in real applications. Pan *et al* [283, 284] performed systematical experiments to investigate the influence of temperature variation on stereo-DIC measurements, and surprisingly found that the temperature-induced strain errors are approximately 30–50 $\mu\varepsilon$ °C^{-1}. This is too large to be neglected in the application of stereo-DIC. To alleviate this influence, preheating is recommended for indoor or short-term outdoor use, and reference specimen compensation for prolonged non-laboratory applications of stereo-DIC.

4. Current state-of-the-art DIC techniques and remaining problems

Being deformation techniques based on digital image capture, image processing and numerical computing, high-quality DIC measurements can be realized only by comprehensively and carefully considering the technical details involved in each step of its practical implementation. Based on the advances made in the past years, important issues involved in optical image capture using regular 2D-DIC and stereo-DIC systems, and key technical details of the state-of-the-art correlation algorithms, are summarized in table 2, in which the superscripts ($*$RP) denote the remaining problems that should be solved in future to realize better DIC measurements.

4.1. Problem #1: speckle pattern fabrication for samples with weak features or in harsh environments

In certain circumstances (e.g. structural testing of aerospace components, displacement monitoring of large engineering structures), it is not allowed or is very difficult or impossible to decorate artificial speckle patterns on the surface of test samples. Without an artificial random speckle pattern, image matching would be much less accurate and may even fail, because the natural texture on the sample surface generally cannot offer sufficient intensity information for accurate displacement tracking, thus posing an important challenge to the DIC technique in dealing with a poor natural texture. Furthermore, in some extreme and/or complex circumstances (e.g. a high-temperature or oxidated environment, high-speed airflow environment, underwater or marine environment, chemical corrosion environment), speckle patterns generally cannot stably adhere to the sample surface or may vary apparently because of complex experimental conditions, such as high temperature, moisture, corrosion, oxidation, etc. Corresponding approaches (using special imaging or new speckle pattern fabrication techniques) that can produce high-contrast, random and stable intensity variations on test sample surfaces in these less than ideal conditions are highly necessary.

4.2. Problem #2: imaging system calibration

Camera calibration is an essential step in using stereo-DIC to link the points of interest on the object surface with their corresponding positions in the left and right cameras. Usually, binocular or multi-camera stereo-DIC imaging systems can be calibrated using a regular-sized calibration target, but difficulties would arise when dealing with large objects (e.g. large engineering structures) or panoramic measurements. In these cases, a regular-sized calibration target is no longer applicable, while with the fabrication of a larger or special calibration target is difficult and expensive to ensure the geometrical quality. Despite several special calibration techniques having been proposed (e.g. a CAD-based [285], a projection-based [286] and a speckle-based calibration method [274]), these methods are not universally applicable. To calibrate the stereo-DIC imaging system when measuring large objects or engineering components, a generalized, easy-to-implement and high-precision camera calibration method is desirable. This matter is expected to be paid more attention in the future.

4.3. Problem #3: adaptive selection of optimal subset size and shape function for subset matching

Subset size and shape function are the two key parameters that should be specified by DIC users when using existing standard subset-based DIC approaches. In ideal cases, the subset size selected for each measurement point should be large enough to comprise sufficient intensity variations to suppress the random error caused by image noise. Also, the corresponding shape function should perfectly approximate local deformation within the target subset. Otherwise, systematic errors will be present due to under-matched shape functions, or increased random errors due to over-matched shape functions. Since the actual deformation occurring in each subset cannot be known in advance in practical applications, an approach is still lacking for the adaptive selection of optimal subset size and shape function for each calculation point, which can comprehensively consider local speckle pattern quality and local complex deformation state.

4.4. Problem #4: fully automatic, robust and fast estimation of initial guess

Although various approaches can be used for the initial guess of seed points, as discussed in section 2.3.4, a fully automatic, robust yet efficient initial guess estimation method, which can accurately estimate six deformation parameters (two displacement components and four displacement gradient components) from target images (even if these images have poor texture, are acquired by different imaging devices, or have undergone large rotation or deformation), has not yet been reported.

4.5. Problem #5: adaptive selection of optimal strain window size and shape function for strain estimation

To extract strain maps from noisy displacement fields consisting of an array of regularly distributed discrete points, the pointwise least squares (PLS) strain estimation algorithm based on local polynomial fitting is most widely used. However, strain window size and the order of fitting polynomials must be defined by the users. Apparently, a larger strain window leads to a better noise suppression effect, with reduced random errors in estimated strain fields. However, since the

Table 2. Summary and comments on the technical details of the state-of-the-art DIC techniques.

DIC implementation procedures	Remarks
Deformation carrier	
■ Speckle pattern:	
Use either natural texture or artificial speckle patterns(*RP1)	*The speckle pattern should be high-contrast, random, isotropic and stable*
Experimental setup	
■ Cameras:	
Use high SNR digital cameras	
■ Illumination and imaging system:	
Use actively illuminated and coupled band-pass filter imaging	*To suppress the adverse effect of ambient variations*
Use cross polarization	*To prevent saturated pixels due to specular reflection*
Use a bilateral telecentric lens or employ the RSC method (2D-DIC)	*To eliminate the errors due to out-of-plane motions and lens distortion for 2D-DIC*
Preheat the cameras or use the reference compensation method (stereo-DIC)	*To mitigate thermal errors due to camera temperature variations*
Algorithm details for displacement field measurement	
■ Imaging system calibration: (*RP2)	
Record at least 50 image-pairs of the calibration target	*Diminish the standard deviation on the calibration parameters*
Feature points detection: combining frontal image and DIC	*To compensate for the imprecision of calibration target fabrication and distortion of lens and perspective*
Camera calibration with bundle adjustment method	
■ Preprocessing of speckle images:	
Gaussian low-pass filtering	*To minimize interpolation-induced bias error*
■ Definition of calculation parameters:	
ROI, seed point, subset size(*RP3), and grid step	
■ DIC analysis:	
Correlation criterion: ZNSSD	*Insensitive to offset and scale intensity changes in the target subsets*
Shape function(*RP3): 1st or 2nd-order shape function	
Subpixel registration algorithm: IC-GN algorithm	*Subpixel registration with enhanced efficiency and better noise robustness*
Interpolation method: Bi-quintic B-spline interpolation	
■ Calculation strategy and initial estimation:	
Calculation strategy: RGDT strategy	*Initial guess of the calculation points can be predicted from the calculated adjacent points*
Initial guess of the seed point(*RP4)	
■ Incremental calculation strategy:	
Update reference image according to the ZNCC coefficient of the seed point	
Use nearest integer subset offset scheme	*To avoid subpixel interpolation of the updated reference subsets*
■ Post-processing of displacement data:	
Threshold according to ZNCC coefficient	
Strain field calculation	
■ Definition of strain calculation parameters:	
Strain window size, and order of the fitting polynomials(*RP5)	
■ Strain calculation strategy:	
Pointwise least squares fitting algorithm	

actual local deformation of a test sample, in general, is complicated and cannot be known beforehand, a mismatch between the adopted polynomials and the underlying local deformation gives rise to either bias error (due to under-matched polynomials) or random errors (because of over-matched polynomials). In many mechanical applications, strain information is often more valuable than displacement data. The accurate estimation of strain distributions from noisy displacement fields for samples subjected to non-homogeneous deformation or strain concentration is undoubtedly a very challenging problem, and necessitates an approach for the adaptive selection of optimal strain window size and local fitting polynomials.

5. Concluding remarks and the future goals

DIC techniques have been widely accepted as a routine experimental practice for full-field displacement and strain measurements in academic research and university laboratories, and demonstrate ever-increasing potential in various engineering fields and industrial applications. In this review, the historical development of the 2D- and stereo-DIC techniques are classified into two phases (i.e. the foundation-laying phase and boom phase), and important advances made to the DIC techniques since 2010 are reviewed. By reviewing the state-of-the-art subset-based DIC techniques, several remaining questions are summarized from the author's personal viewpoint.

In the experimental mechanics community, DIC has been considered as a revolutionary advance; the most important advance since the strain gauge. It will continue to be the most practical and powerful deformation measuring tool for the foreseeable future. There is no doubt that the powerful and versatile DIC techniques will find more and more applications in various new areas in the future. To further advance research interest and better application of DIC techniques, possible developments that could make DIC a more useful and incisive technique are listed here.

1. An intelligent, fully automatic, high-accuracy and high-efficiency DIC framework should be developed, that integrates all of the following aspects:

   – Provide accurate estimation of the initial guess of seed points in an automatic and fast manner.
   – Adaptively select optimal subset size and shape function for accurate displacement field measurements.
   – Adaptively select optimal strain window size and fitting polynomials for accurate strain field determination.

It should be noted that the standardization of DIC has been a fashionable concept in the DIC community. However, one prominent advantage of DIC techniques is its flexibility and versatility. Thus, considering the diverse imaging systems used, and the distinctly different measurement objects and conditions, it is difficult or even impossible to develop standards generally applicable to each individual application. However, from the author's personal perspective, standardization of the DIC algorithm can be implemented provided the challenging problem of the adaptive selection of calculation parameters is solved.

2. Introduction of new imaging systems, and imaging error quantification and correction:

   – Combine new imaging devices (e.g. panoramic cameras, plenoptic cameras, endoscopes, ultrasonic electrography, photoacoustic imaging, optical coherence tomography, neutron imaging, super-resolution imaging) with DIC to offer new tools for quantitative deformation or high-throughput (i.e. multiple physical quantities) measurements in new areas.
   – Understand the imaging principle and imaging model of new imaging devices, identify and quantify their error sources, and develop approaches to mitigate or correct these error sources to provide reliable and accuracy deformation measurement.

3. Integration with modern design and simulation tools, and inverse methods:

   – Combine DIC with CAD tools and FEM software to obtain an integrated suite of design, analysis and measurement tools.
   – Integrate DIC techniques with various inverse methods (e.g. virtual fields method, finite element model updating) to realize seamless and accurate identification of multiple constitutional parameters.

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